Supplementary Material: Humans Are More Diverse: Frontier LLMs Show Extreme Policies in Idealised AI Development Races

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Submission Id: TBD********†

ABSTRACT

This document holds the material the main paper points to but has no room to carry: the full breakdown of every task-validity probe, the replication check against the human benchmark, a claim-by-claim audit of the citations, the persona and measured-risk comparison, the predictive analysis behind the reported drivers, and the complete descriptive baselines for two-player and N-player races. Nothing here is needed to follow the main argument. Everything here comes from the same runs, under the same admission rules, and is reported at the same evidence level.

KEYWORDS

Large language models, AI development races, multi-agent systems, repeated games, AI safety, task validity

HOW TO READ THIS DOCUMENT

Each section stands on its own and is named for what it measures. Section letters restart at A because this is supplementary matter.

- **Task-validity probe battery.** Every probe, its wording family, and the per-route pass rate behind the summary in the main paper.
- **Nine-route endpoint-admission campaign.** The full per-domain admission result for every route the main paper reports, including the two domains it omits, the admission thresholds, and the retained failure.
- **Human-benchmark replication check.** Whether the reference human study's reported pattern reproduces on the released participant data under our own analysis code.
- **Claim-scoped citation audit.** For each cited work, the claim it is used to support and the claim it does not support.
- **Risk preference and assigned personas.** The persona families, and how a prompted persona compares with measured human risk preference.
- **Selection-strength sweep of the evolutionary benchmark.** Whether the gap between theory and the audited routes depends on one parameter setting.
- **Crossed representation-robustness cell.** The one narrative skin by symbol mapping rerun that completed, and why no general mapping claim follows from it.
- **Predictive structure of unsafe choices.** The feature analysis behind the drivers reported in the main paper.
- **Two-player descriptive baseline, full scope.** Every checkpoint and condition, including those the main paper does not plot.
- **N-player self-play scope.** Group sizes three to five.
- **Trajectory embedding, by population and by outcome.** The same embedding twice, coloured by population and then by UNSAFE rate.
- **Cluster-count sensitivity and trajectory diversity.** How the human archetypes and the diversity comparison behave as the analysis choices move.
- **UNSAFE rate by persona band and relative position.** Cells reported only here, with observed rather than assigned position.

Four terms are used with fixed meanings throughout. A *route* is the exact provider string a request is addressed to, for example `google/gemini-3-flash-preview`. An *endpoint* is a route that carries an admission verdict from the probe battery below. A *checkpoint* is one dated model version, which is what a comparison across the rows of a behavioural table compares. *Model* is used only generically, for a language model as a kind of system. Where a number belongs to a route rather than to a model in general, the route string is printed beside it.

A EXTENDED RESULTS AND ROBUSTNESS CHECKS

This supplement collects the material referenced from the results section in the main paper as supporting rather than extending the main claims: the full task-validity probe breakdown (§A.1), the human-benchmark replication check (§A.8), the persona comparison (§A.10), the predictive analysis (§A.12), and the full two-player and N-player descriptive results (§A.13, §A.14). It also carries the evidence the main paper only summarises: the nine-route admission campaign (§A.3), the selection-strength sweep of the evolutionary benchmark (§A.6), the one completed crossed representation-robustness cell (§A.7), the matched group-size comparison (§A.22), the scripted-opponent campaign (§A.23), the cluster-count sweep behind the human archetypes (§A.19), and the trajectory-diversity statistics (§A.20).

A.1 Task-validity probe battery: full breakdown

Table 1 reports the full breakdown behind the first confirmatory frontier endpoint admission, protocol v5, which admitted two routes. It is superseded as the campaign the main paper reports by the nine-route v6 campaign of §A.3, and is retained here because no v6 verdict contradicts it. The confirmatory audit used the two exact routes listed in Table 2 under the frozen protocol v5: 20 atomic probes spanning rule recall, one-stage payoffs, state reconstruction, state transition, terminal scoring, and expected payoff, repeated three times per route, for 60 retained outputs per route and 120 retained outputs in total. Every response was schema-valid and no transport retry was needed. Numeric probes used a fixed structured-output contract; categorical probes explicitly listed the allowed answer set.

The largest errors were in the optional payoff check: three of six Gemini responses and none of the Claude responses were correct. We retain this domain as a diagnostic. Live gameplay asks for an action, whereas the state and terminal domains determine whether

Table 1: Task-validity probe accuracy by category for the two admitted frontier routes, protocol v5 (n = 60 retained probe outputs per route).

Probe category	Gemini 3 Flash	Claude Sonnet 5
Rule recall	100.0%	100.0%
One-stage payoff lookup	100.0%	100.0%
State reconstruction	93.3%	86.7%
State transition	100.0%	100.0%
Terminal scoring	100.0%	80.0%
Expected-payoff calculation	50.0%	0.0%
Overall semantic accuracy	*93.3%*	*81.7%*
Schema-valid responses	*100.0%*	*100.0%*

the action prompt is mechanically interpretable. The admission task therefore records both the required gate and the arithmetic limitation instead of silently dropping the endpoint.

A.2 Frontier endpoint registry and admission outcomes

Endpoint availability was checked against the active Kaggle Benchmark identity and recorded using the exact route string. One repetition of the 20-probe bank was used as a diagnostic smoke for every advertised route; the two routes that passed the v5 gameplay thresholds then received the three-repetition confirmatory admission. A route that was unavailable or failed transport was retained as failed infrastructure evidence, never replaced by another endpoint.

Table 2: Per-endpoint frontier admission audit. Smoke results are diagnostic; only routes with a completed v5 artifact and passing gameplay thresholds enter the new gameplay run.

Exact route	Overall	State	Terminal	Status
`google/gemini-3-flash-preview`	93.3%	93.3%	100%	admitted
`anthropic/claude-sonnet-5@default`	81.7%	86.7%	80%	admitted
`google/gemini-3.1-flash-lite-preview`	75%	60%	80%	diagnostic
`openai/gpt-5.4-nano-2026-03-17`	50%	20%	60%	diagnostic
`deepseek-ai/deepseek-r1-0528`		not admitted		transport failed
`ibm/granite-4.0-h-small`		not admitted		transport failed
`openai/gpt-oss-120b`		not admitted		transport failed
`qwen/qwen3-next-80b-a3b-instruct`		not admitted		transport failed

The required gameplay gate was overall accuracy at least 80%, state reconstruction accuracy at least 75%, and terminal-scoring accuracy at least 75%. Expected-payoff accuracy was reported separately because it is a closed-form arithmetic diagnostic rather than an action-task requirement. The active identity exposed no second authorised account, so no multi-account pooling is claimed.

A.3 Nine-route endpoint-admission campaign

The main paper summarises a later and larger admission campaign, `ai-race-frontier-admission-v6`, which put nine routes through the same gate. This subsection gives it in full, including the two domains the main paper's table omits for space: rule recall and one-stage payoff lookup. Table 3 is the complete per-domain result. Each route answered the same frozen bank of 20 probes three times, giving 60 retained rows per route and 540 retained rows in total, at temperature 0 with a 256-token output cap and base seed 260726, on the Kaggle Benchmark identity `daosyduyminh`.

The three admission thresholds were fixed before the campaign ran: overall semantic accuracy at least 80%, state-reconstruction accuracy at least 75%, and terminal-scoring accuracy at least 75%. A route is admitted only if it clears all three. Expected-payoff accuracy is recorded but never blocks admission, because it is a closed-form arithmetic diagnostic rather than something the action task requires. Five of the nine routes were admitted: `google/gemini-3-flash-preview`, `anthropic/claude-opus-5@default`, `openai/gpt-5.4-2026-03-05`, `openai/gpt-5.5-2026-04-23`, and `anthropic/claude-sonnet-5@default`. Four were refused: `google/gemini-3.1-flash-lite-preview` on state reconstruction alone, `openai/gpt-5.4-mini-2026-03-17` on overall accuracy and state reconstruction, and `google/gemini-3.5-flash-lite` and `openai/gpt-5.4-nano-2026-03-17` on all three thresholds.

Three facts make the nine rows of Table 3 directly comparable with one another. All nine ran under the same protocol identifier, `ai-race-frontier-admission-v6`. All nine answered the same probe bank, whose SHA-256 hash is this.

```
2953fb472dcba47511de108a47d4126b
0f661e79bf60c96a3ab46f581abca8bc
```

The shared hash is what licenses placing the routes side by side, and the analysis script fails closed, marking a route non-comparable and refusing to tabulate it beside the others if its hash differs. All nine also read the same rules context, whose hash is this.

```
b80981a5e888d04649300f79b7388ca7
a755b958f239f0172efd9ca35aab2ae0
```

Eight routes ran at benchmark task version 7 and one, `google/gemini-3.5-flash-lite`, at task version 8.

That one route's decoding contract differs from the other eight, and the difference is visible in its manifest rather than inferred. The eight version-7 routes were sent an explicit reasoning budget of `none`. The version-8 route omits the reasoning-budget argument altogether, because this provider rejects the presence of the argument itself, returning an HTTP 400 invalid-argument error before a single probe is sampled. That is a mismatch in the transport contract, not evidence about the model, and it had previously removed the route from the audit entirely. The amendment that introduced the per-route contract is dated 2026-09-09 in `docs/reviewer-revision-frontier-protocol.md`, which also records its scope: for every route whose resolved contract is still `none` the outgoing request is byte-identical to the earlier runs, so those artifacts stay poolable with them, while a route whose resolved contract omits the argument records that omission in its manifest, and any table that pools such a route must say so. This table says so.

One failed run is retained rather than deleted. Run 1494041, route `google/gemini-3.5-flash-lite` at task version 7, ended in a `RuntimeError` from exhausted bounded transport retries. It appears nowhere in Table 3 because it produced no scoreable probe answer, and it is kept in the accounting chain as a transport failure under `results/frontier/admission_campaign_v6/`

Table 3: Complete nine-route endpoint-admission result, protocol `ai-race-frontier-admission-v6`. Every cell is the percentage of that domain's probes answered correctly, out of 60 retained probe answers per route (20 probes, three repetitions each). "Task ver" is the frozen benchmark task version the route ran under. Expected payoff is a recorded diagnostic and does not gate admission; the three gates are overall accuracy at least 80%, state reconstruction at least 75%, and terminal scoring at least 75%. Rule recall, stage payoff, state transition, and expected payoff each rest on six probe answers per route, so their accuracy can only move in steps of one sixth, about 16.7 percentage points.

Exact route	Task ver	Rule recall	Stage payoff	State recon.	State trans.	Terminal scoring	Exp. payoff (diagnostic)	Overall	Admitted
`google/gemini-3-flash-preview`	7	100.0	100.0	93.3	100.0	100.0	50.0	93.3	yes
`anthropic/claude-opus-5@default`	7	100.0	100.0	93.3	100.0	100.0	33.3	91.7	yes
`openai/gpt-5.4-2026-03-05`	7	100.0	100.0	100.0	100.0	100.0	0.0	90.0	yes
`openai/gpt-5.5-2026-04-23`	7	100.0	100.0	80.0	100.0	100.0	50.0	90.0	yes
`anthropic/claude-sonnet-5@default`	7	100.0	100.0	100.0	100.0	80.0	0.0	85.0	yes
`google/gemini-3.1-flash-lite-preview`	7	100.0	100.0	73.3	100.0	80.0	16.7	80.0	no
`openai/gpt-5.4-mini-2026-03-17`	7	100.0	100.0	66.7	66.7	86.7	0.0	75.0	no
`google/gemini-3.5-flash-lite`	8	100.0	100.0	60.0	50.0	60.0	0.0	65.0	no
`openai/gpt-5.4-nano-2026-03-17`	7	75.0	100.0	20.0	50.0	66.7	0.0	51.7	no

`failed_runs/`. The successful rerun of that same route is the task-version-8 row.

No route changed its verdict relative to the earlier v5 campaign of §A.1 and §A.2, which is the only earlier campaign entitled to admit anything. One reason for refusal did change: `google/gemini-3.1-flash-lite-preview` rose from 76.7% to 80.0% overall accuracy and now clears the overall gate, yet is still refused, because its state reconstruction is 73.3%, below the 75% threshold. Expected payoff is the weakest domain on every one of the nine routes, including the strongest, which is why it was frozen as a diagnostic before the campaign began: had it gated admission, no route would have been admitted at all.

A.4 Frontier neutral baseline

The new neutral gameplay baseline used protocol v3, the same mechanism and prompt for both admitted routes, ten repetitions per risk level, and 30 races per route. Each route produced 558 structured decisions with zero parse failures. The entries in Table 4 are decision-weighted UNSAFE rates; they are not pooled across routes and do not estimate an evolutionary equilibrium.

Table 4: Neutral two-player frontier baseline under the admitted routes. Each route has 30 races, 558 decisions, and zero parse failures.

Exact route	$r = .1$	$r = .6$	$r = .9$	Races
`google/gemini-3-flash-preview`	98.9%	74.2%	59.1%	30
`anthropic/claude-sonnet-5@default`	89.2%	46.2%	37.6%	30

The Gemini manifest records that the hosted SDK stripped the requested sampling seed. The Claude manifest records that the seed was forwarded but provider-side application was not confirmed. Temperature was requested as 0.7 for both routes but was not forwarded by this SDK path. These facts bound the result to the exact hosted route and software contract.

To make the comparison boundary explicit, Figure 1 separates the risk response, the route-specific departure from the strong-selection reconstruction, the nearest-rule diagnostic, and its mismatch rate. This is an analysis artifact rather than an equilibrium test: the frontier routes generate prompted self-play trajectories, whereas EGT samples stationary population composition.

The invasion view in Figure 2 exposes the selection topology behind the three theoretical endpoints. Its directed edges are computed by the archive-compatible EGTtools implementation. The node areas use the finite-mutation main-reference chain because the small-mutation limit can contain multiple absorbing classes at extreme payoff scales. This is a diagnostic decomposition, not a claim that the frontier routes sample the same population process.

A.5 How much of a route's number is the run?

Every route-level rate in this paper comes from one thirty-race run, so a reader is entitled to ask how much of a rate is the route and how much is the particular run. One route answers that question directly. Sampling on these endpoints is not reproducible: the SDK strips the seed request before it reaches the provider, which every run manifest records as `not_applied_sdk_stripped_for_route`. Two runs of the same cell are therefore genuinely independent draws from the route, not a rerun of one draw.

`google/gemini-3-flash-preview` was collected twice under the frozen protocol `ai-race-frontier-baseline-v3`: at task version 3 on 2026-09-08, which is the run reported everywhere else in this paper, and again at task version 4 on 2026-09-09 on a different billing identity. The repeat was not planned as evidence. It rode along with the push that carried the reasoning-budget amendment of §A.3 and was recovered from the Kaggle archive afterwards. Protocol, prompt hash, mechanism, seed structure and the full set of thirty (game seed, risk) blocks are identical across the two runs, and `scripts/analyze_baseline_replication.py` refuses the comparison unless they are.

Agreement is close. UNSAFE rates move from 98.9% to 100.0% at $p_r^{\max} = 0.1$, from 73.1% to 74.2% at 0.6, and from 60.2% to 59.7% at 0.9: the largest per-risk difference over 186 decisions is 1.1 percentage points, and the risk response, the quantity the admission analysis in §A.3 actually uses, moves from 38.7 to 40.3 percentage points. That is small beside the between-route spread of 7.0 to 100.0 points that the same analysis reports, which is the comparison this check exists to protect.

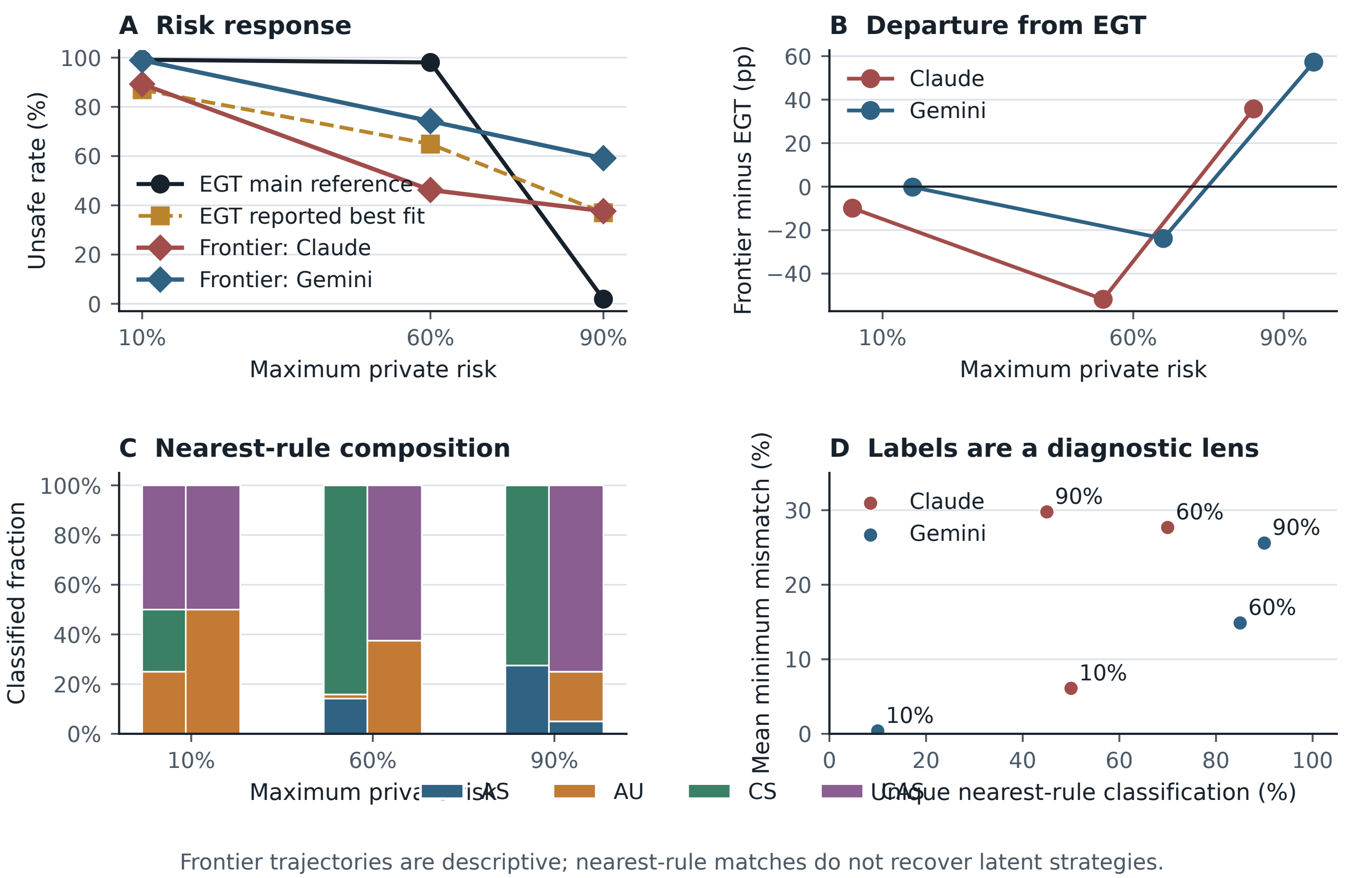


Figure 1: Boundary analysis for the reconstructed evolutionary model and the two admitted frontier routes. Panel A shows route-specific UNSAFE responses alongside the two EGT parameter regimes. Panel B shows the deviation from the strong-selection EGT reference. Panel C gives the nearest-rule composition, and Panel D shows why those labels are only a diagnostic lens: uniqueness and minimum mismatch vary across routes and risk levels.

It is one route and one repeat, so it bounds run-to-run variation for this route at this sample size and licenses nothing for the routes that were collected once. The repeat is stored at `results/frontier/baseline_replication/`, deliberately outside the campaign tree: both campaign analysers key their results by model route, so a second run of an already-represented route placed inside that tree would not raise an error, it would quietly displace the reported one.

A.6 Selection-strength sweep of the evolutionary benchmark

The comparison in §A.4 uses one setting of the reconstructed evolutionary model. A reviewer may reasonably ask whether the gap between that model and the audited routes is a property of the model or an artifact of that one setting. This subsection answers that question by sweeping the setting.

Two numbers govern the model. *Selection strength* β says how strongly a simulated agent prefers to copy a better-performing neighbour: at β near zero copying is almost random, and at large β it is almost deterministic. The *mutation rate* μ is the chance, per update, that the sampled individual abandons its strategy and switches uniformly to one of the other three. The sweep covers ten values of selection strength, $\beta \in \{0.001, 0.003, 0.01, 0.03, 0.1, 0.3, 1, 2, 3, 10\}$, crossed with three rules for setting the mutation rate: $\mu = \beta/Z$, the main-text convention, which gives $\mu = 0.02$ at $\beta = 2$; $\mu = 1/Z = 0.01$; and a fixed $\mu = 0.05$, the convention under which the source study reports its best fit. With $Z = 100$ individuals, four strategies, and the mechanism's three maximum-private-risk levels (0.1, 0.6, 0.9), the sweep contains 90 cells built from 29 distinct parameter pairs, since some pairs coincide across rules and were simulated once. Each cell is estimated from four independent chains of 400,000 steps with a burn-in of 100,000, thinned by 100, giving 4,000 retained samples per chain.

The reference setting reproduces. At $\beta = 2$ with $\mu = \beta/Z = 0.02$, the sweep returns a decision-weighted UNSAFE share of 99.2%, 98.0%, and 1.9% at maximum risks 0.1, 0.6, and 0.9: the same three-point profile the main paper attributes to the published reference, matched to within 0.0002, 0.0001, and 0.0001 in share, far inside the 0.02 tolerance the check declares in advance. The sweep uses a

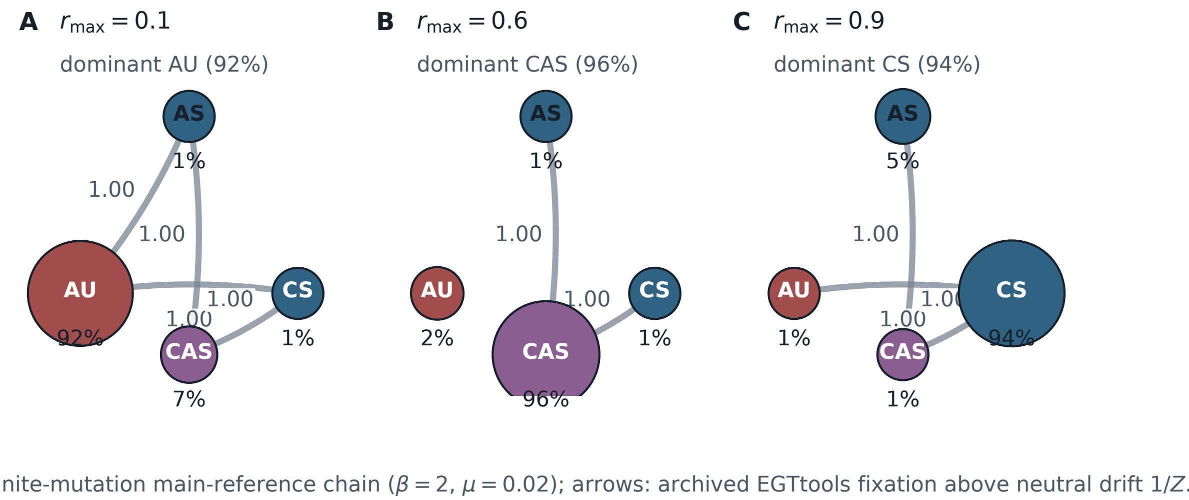


Figure 2: Risk-conditioned invasion topology for the reduced evolutionary game. Node area shows the finite-mutation main-reference chain share ($\beta = 2$, $\mu = 0.02$); directed edges show EGTtools `PairwiseComparison` fixation probabilities above neutral drift. The figure is a hybrid diagnostic: the edge topology is evaluated in the EGTtools small-mutation limit, while node areas come from the independently simulated finite-mutation chains. It should not be read as a latent-strategy diagram for the language models.

deliberately disjoint seed scheme from the reproduction script, so this agreement is an independent-seed check rather than a rerun of the same random numbers.

The qualitative gap does not close anywhere on the grid at the reference selection strength, but it does close at very weak selection. Of the 90 cells, four reproduce the qualitative shape the admitted endpoints show, namely near-total UNSAFE play at risk 0.1, a middling rate at 0.6, and a lower but clearly non-zero rate at 0.9, declining at every step. Three of those four also pass the mixing diagnostic. All of them sit at weak selection: $\beta = 0.01$ and $\beta = 0.03$ under $\mu = 0.05$, and $\beta = 0.003$ and $\beta = 0.01$ under $\mu = 1/Z$. Ten of the 90 cells are flagged as mixing-suspect, meaning their four independent chains disagreed by more than 0.05 on the aggregate UNSAFE share, so their chain mean must not be read as a stationary distribution; almost all of them are the very small mutation rates that $\mu = \beta/Z$ produces at small β, which make the four uniform-strategy states nearly absorbing.

The best-fitting cell is the same for both audited routes, and it is not the reference setting. Fitting is by root-mean-squared deviation, in percentage points, between the model's three-point UNSAFE profile and the route's observed decision-weighted UNSAFE rates at the same three risk levels; the largest single-condition deviation is reported beside it so that one badly missed condition stays visible. For both routes the minimum-error cell is $\beta = 0.01$ with fixed $\mu = 0.05$, whose profile is 87.3%, 63.9%, and 38.0%. Against `anthropic/claude-sonnet-5@default`, observed at 89.2%, 46.2%, and 37.6%, that cell has an error of 10.28 percentage points and a largest single-condition miss of 17.70 points, at risk 0.6. Against `google/gemini-3-flash-preview`, observed at 98.9%, 74.2%, and 59.1%, the error is 15.13 points and the largest miss 21.14 points, at risk 0.9. Even the best cell on the grid is therefore more than ten percentage points away from either route.

One boundary is worth restating, because the reproduction lane states it itself. This is a repo-native reconstruction of the disclosed reduced evolutionary model, not a bitwise reproduction of the original authors' code: that code, the generated payoff matrices, the payoff Monte Carlo seeds, and the exact library revision are not public in the preprint version we worked from, and the compiled comparison class could not be executed on this interpreter because no matching wheel exists for it, so the sweep ran the pinned pure-Python implementation instead. The quantities above are therefore seeded-chain estimates reported with their between-chain spread, not exact stationary distributions, and the route rates beside them are descriptive prompted self-play endpoints rather than draws from a population process.

A.7 Crossed representation-robustness cell

An earlier local pilot varied the story the game is told through and the symbols used for the two actions, but varied them together with repetition parity, so the two could not be separated, and it failed its comprehension gate. The main paper therefore excludes it and requires a fully crossed rerun before any claim about representation. That rerun has now completed on both admitted routes, and this subsection reports it.

The rerun crosses two *narrative skins*, that is two stories that leave the rules untouched, against two *symbol mappings*, that is which of the two opaque response codes P and Q stands for SAFE and which for UNSAFE, inside repetition blocks, at each of the mechanism's three maximum-private-risk levels. That is twelve cells per route. It completed on both admitted routes under protocol `ai-race-frontier-context-mapping-v3`, each with ten repetitions, 120 races, 2,232 recorded decisions and zero parse failures: Gemini 3 Flash as run 1373757 and Claude Sonnet 5 as run 1504455,

Table 5: Fully crossed narrative-skin by symbol-mapping rerun on the single route that completed it, `google/gemini-3-flash-preview`. "Mapping" names which response code stands for SAFE: under `P_SAFE_Q_UNSAFE` the code P means SAFE, and under `Q_SAFE_P_UNSAFE` it means UNSAFE. n_{race} is the number of independent races in the cell and n_{dec} the number of decisions recorded in it; the UNSAFE rate is decision-weighted. All twelve cells are complete and no response failed to parse.

Narrative skin	Mapping	Risk	n_{race}	n_{dec}	UNSAFE actions	UNSAFE rate
abstract_game	P_SAFE_Q_UNSAFE	0.1	10	186	186	100.0%
abstract_game	P_SAFE_Q_UNSAFE	0.6	10	186	156	83.9%
abstract_game	P_SAFE_Q_UNSAFE	0.9	10	186	111	59.7%
abstract_game	Q_SAFE_P_UNSAFE	0.1	10	186	186	100.0%
abstract_game	Q_SAFE_P_UNSAFE	0.6	10	186	186	100.0%
abstract_game	Q_SAFE_P_UNSAFE	0.9	10	186	166	89.2%
technology_race	P_SAFE_Q_UNSAFE	0.1	10	186	186	100.0%
technology_race	P_SAFE_Q_UNSAFE	0.6	10	186	144	77.4%
technology_race	P_SAFE_Q_UNSAFE	0.9	10	186	81	43.5%
technology_race	Q_SAFE_P_UNSAFE	0.1	10	186	186	100.0%
technology_race	Q_SAFE_P_UNSAFE	0.6	10	186	185	99.5%
technology_race	Q_SAFE_P_UNSAFE	0.9	10	186	83	44.6%

whose full route strings are given in Table 2. The horizon and set-back random-number streams are held common across conditions, so two cells differ in their prompt and not in the environment they met. Table 5 gives all twelve cells for the first route; the second route's cells are in the derived table named at the end of this subsection, and both routes' paired contrasts are in Table 6.

Pooling cells is not the sharpest way to read this design. The repetition index is a common-random-number block, so the same index reuses the same sampled horizon across every skin and mapping cell, and the recorded `horizon_draws_sha256` lets that be checked rather than assumed. It holds for all 60 paired blocks on both routes. Differencing inside a repetition therefore removes the horizon draw from the comparison and leaves the presentation as the only thing that varies. Table 6 reports those paired contrasts, with intervals from a percentile bootstrap over repetition blocks.

Two things stand out. The symbol mapping moves play by a similar amount on both routes, about eight to ten percentage points, even though the two routes differ by provider and sit six points apart on the admission audit. The narrative skin also moves play on both, but by very different amounts, so skin sensitivity is route-specific in a way that code sensitivity is not. All four intervals exclude zero. Nothing in the mechanism differs across any of these cells: the payoff matrix, the progress increments, the hidden horizon law and the risk formula are identical, and both routes had already passed the comprehension gate. Two routes is enough to show the effect replicates and not enough to characterise endpoints in general.

Both routes' marginal cells and both paired contrasts are regenerated by `analyze_frontier_context_mapping_cross.py` under `scripts/`, which refuses a route whose race count, ten-per-cell balance or parse-failure count does not check out. Its outputs are `context_mapping_cross_marginals.csv` and a provenance `.json` beside it. An earlier attempt on the Claude route stopped at 106 of 120 races on a provider quota refusal; it is retained as a superseded failure record with a pointer to the completed run, and none of its cells is reported, because a counterbalanced design read at 106 of 120 races is unbalanced rather than merely smaller.

Table 6: Repetition-paired presentation contrasts on the two crossed routes. Each contrast is the mean difference in UNSAFE rate within a repetition block, which holds the sampled horizon fixed; positive means the second level raises UNSAFE play. Intervals are 95% percentile bootstrap intervals over the 60 paired blocks. Horizon pairing was verified for 60 of 60 blocks in every row.

Route	Contrast	Effect	95% CI
Gemini 3 Flash	code Q means SAFE	+9.5 pp	+5.6 to +13.8
Claude Sonnet 5	code Q means SAFE	+8.3 pp	+5.4 to +12.2
Gemini 3 Flash	abstract contest	+10.8 pp	+6.0 to +15.6
Claude Sonnet 5	abstract contest	+3.4 pp	+0.9 to +7.4

Two things are visible in Table 5 and neither is a general finding. First, the risk response survives both manipulations: in every one of the four skin-by-mapping combinations the UNSAFE rate is 100% at risk 0.1 and lower at risk 0.9. Second, the mapping is not inert. Holding the skin fixed, swapping which code means SAFE moves the rate at risk 0.9 from 59.7% to 89.2% under the abstract story and from 43.5% to 44.6% under the technology story, and at risk 0.6 from 83.9% to 100% and from 77.4% to 99.5% respectively. A relabelling that a correct model of the game should ignore therefore changes the action sequence.

No general claim about symbol mapping or narrative skin follows from this table, because the design is crossed for one route only. A single route cannot separate a property of this task from a property of this provider's decoder at this revision, and a second route is what would be needed to do so. The matched rerun on `anthropic/claude-sonnet-5@default` did not complete: it was stopped by a provider quota refusal, and a run stopped part-way is a failure record in this project's accounting, not a partial result to be reported beside a complete one. We therefore report the completed cell, name its single-route limit, and make no mapping claim.

A.8 Human-benchmark replication check

Before comparing model-generated behaviour with the human benchmark of Domingos and Han [4] in the two-player strategic diversity results in the main paper, we first validated our reconstruction of that benchmark by refitting its published dynamic specification on the same public de-identified data. The reconstructed estimation sample contains 2,888 round-level observations, 172 race-pair clusters, and 338 participants. Table 7 compares the refit coefficients against the values reported in the original study.

Four human sample sizes appear across this supplement, and they differ because each analysis needs a different part of the same public record. The raw first-five-round sequence comparison keeps the 340 participants with a complete five-round action history. The descriptor and elicitation analyses keep 341, because the five aggregate descriptors and the elicited risk score can be computed for one further participant whose complete sequence is unavailable. The dynamic refit above keeps the 338 participants with complete values on every covariate in the published specification. No participant is added anywhere; each figure is a subset of the same released dataset.

Table 7: Replication of Domingos and Han [4]'s dynamic specification on the public de-identified human data. "Reconstructed" is our refit; "Original" is the coefficient reported in the source study.

Term	Reconstructed	Original
Opponent's previous action	0.606	0.607
Player's progress gap	−0.295	−0.296
First-round action	0.217	0.217
Player's own previous action	−0.195	−0.193

A.9 Claim-scoped citation audit

Table 8 records both the supported use of each cited study and the nearest excluded overclaim. These evidence boundaries do not turn the diagnostic pilots in this paper into confirmatory results.

The reconstruction reproduces the original coefficients almost exactly (largest discrepancy 0.002), which is the basis for treating the human-side comparisons in the two-player strategic diversity results in the main paper and the persona and measured risk results in the main paper as resting on a correctly processed benchmark rather than on a reimplementation error.

A.10 Risk preference and assigned personas

We finally distinguish three constructs that should not be conflated: the human participants' pre-game elicited risk preference, the mechanism's assigned private-risk treatment, and the narrative risk persona supplied to an LLM in its prompt.

The three persona framings themselves are drawn in Figure 3, which the main paper's design section points to rather than reproducing.

Human participants completed an incentivised Eckel–Grossman gamble elicitation (0 = most risk-averse, 5 = least risk-averse) before playing the two-player race: a choice among six gambles, made once, before any round is played. This elicited measure has essentially no direct association with subsequent UNSAFE rate,

$$\text{slope} = -0.004, \qquad r = -0.015, \qquad p = 0.79, \qquad n = 341,$$

where the slope is the trend across the six group means plotted in Figure 4; the participant-level slope of the same relationship is −0.002, while the correlation, the p value and the sample size above are participant-level. The difference in mean UNSAFE play between the most and the least risk-accepting human groups, about five percentage points, lies inside the corresponding uncertainty interval. This null result independently reproduces Domingos and Han's own preregistered finding that elicited risk preference did not significantly predict UNSAFE choice [4], and stands in contrast to the tournament literature's general finding that human risk-taking responds to competitive standing rather than remaining a fixed trait [5, 7, 8]: in this task, the round-by-round interaction captured in the two-player strategic diversity results in the main paper is a stronger organising signal than static pre-game disposition. The multiplayer position analysis is exploratory and is not used as a second independent confirmation of this two-player result.

One further place remained where an association could have survived, and it does not. The elicitation might have failed to predict *how much* UNSAFE a participant plays while still predicting *which* dynamic pattern that participant follows. It does not do that either. Comparing the elicited score across the four behavioural archetypes of §A.19, which are the k-means groups fitted to five standardised trajectory descriptors of the same human participants, returns a null:

$$H = 4.540, \qquad \text{df} = 3, \qquad p = 0.209, \qquad n = 341,$$

by a tie-corrected Kruskal–Wallis test, which asks whether the distribution of one variable differs across several groups without assuming a normal distribution. The effect size is $\varepsilon^2 = 0.0046$, so the archetype label accounts for under half of one percent of the variance in the elicited score, which is negligible on any convention. No participant is excluded: all 341 human participants carry both an archetype label and an elicited score, and the elicitation is complete.

The archetype medians are identical. All four groups have a median elicited score of 2.0 on the 0 to 5 scale, with interquartile ranges of 1.0 to 2.5 ($n = 19$), 1.0 to 3.0 ($n = 170$), 1.0 to 3.0 ($n = 133$), and 1.5 to 3.5 ($n = 19$). None of the six pairwise comparisons between archetypes is significant after Holm–Bonferroni correction across all six pairs. The smallest uncorrected p is 0.079, between the two largest archetypes, which becomes 0.476 after correction; every other corrected p is above 0.88. The same conclusion follows from Dunn's test on mean ranks and from pairwise Mann–Whitney tests, which the artifact reports side by side.

Correction. An earlier version of this supplement reported $H = 21.95$, $p = 0.001$, $n = 286$ for this comparison and read it as a weaker, second-order association surviving the null above. That value has no source anywhere in the analysis record, and no subset of 286 participants arises at any documented step of this pipeline: the elicitation is non-missing for all 341 participants and the clustering drops nobody. It is superseded by the artifact `elicited_risk_by_archetype.json`, whose numbers are the ones reported above.

The null is not a weakness in the comparison; it sharpens it. A measured human risk preference predicts neither the level of UNSAFE play nor the pattern of it, whereas a prompted risk persona moves a checkpoint's UNSAFE rate by between 50.5 and 98.3 percentage points from its lowest to its highest level (Table 9). An elicited disposition and an instructed persona are therefore not two measurements of one construct; they are different kinds of variable, and the shared axis in Figure 4 aligns them for visual comparison only.

Figure 4 (left) shows how little that null leaves. Human mean UNSAFE play stays inside a 49–62% band across the entire elicitation range, never approaching the low-UNSAFE checkpoints of the two-player strategic diversity results in the main paper (GPT-5-nano 17%, Claude Sonnet 5 22%, Claude Opus 5 33%) or the high-UNSAFE Gemini models (73–83%). Elicited risk preference does not move a participant far enough to resemble a different checkpoint's policy.

The right panel is the opposite (Table 9): every checkpoint with a six-level sweep climbs 50.5 to 98.3 points from R1 to R6, all but the two nano checkpoints monotonically, and those two dip by well under a point at R2 before rising. The label does not, however, move checkpoints by a common factor. GPT-5.6 Terra runs lowest at R1 and finishes within 1.3 points of the top, while GPT-5-nano starts fifth of seven and ends last, so the order of the curves is not preserved, and the same instruction is worth very different amounts

Table 8: Source-level audit for citations added to the prompt-sensitivity and behavioural-validity argument.

Source	Supported use	Excluded overclaim
Sclar et al. [12]	Meaning-preserving few-shot formatting changes can yield large, model-dependent performance spreads.	Not evidence for semantic paraphrase effects or a universal effect size.
Pezeshkpour and Hruschka [9]	Reordering multiple-choice answer options can change predictions.	The proposed uncertainty-plus-placement mechanism is not causally identified.
Wei et al. [13]	Order and response-token identity are distinct selection-bias targets.	Not proof that every task or model exhibits the same bias.
Salinas and Morstatter [11]	Controlled output-format and atomic whitespace variants can alter answers.	Not a claim that every added space changes behaviour.
Zheng et al. [15]	Across their factual-question setup, average persona benefit was absent or slightly negative and reliable persona selection remained difficult.	Not a universal null effect for persona-driven agent tasks.
Aher et al. [1]	Three of four behavioural findings were qualitatively reproduced, while the fourth showed a hyper-accuracy distortion.	Not evidence that LLM samples are interchangeable with human populations.
Akata et al. [2]	Prediction before action improved coordination and scores in repeated games.	Not a general claim that an observed action reveals a correct opponent belief.
Herr et al. [6]	Action-label order and payoff reassignment changed choices in two canonical two-player games.	Not evidence that every strategically equivalent transformation has the same effect.
Robinson and Burden [10]	Procedurally varied vignettes produced substantial contextual variability under one underlying Prisoner's Dilemma structure.	Not a repeated-game result and not an estimate for the present race.
Wu et al. [14]	Performance declined consistently across 11 tasks when default mappings were replaced by specified counterfactual alternatives.	Not proof that the tested models lacked all abstract reasoning ability.
del Rio-Chanona et al. [3]	LLM markets recovered broad feedback-condition dynamics but displayed less strategy heterogeneity than human participants.	Not evidence that LLM agents and human participants are exchangeable.

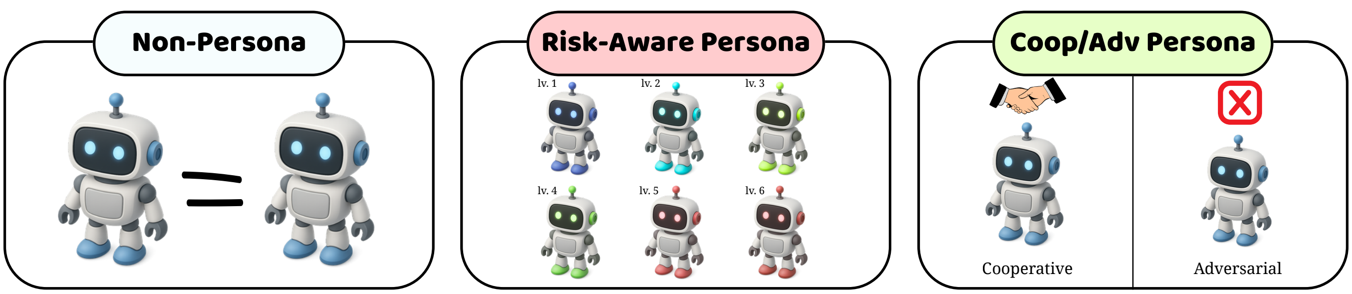


Figure 3: The persona framings assigned to each company's executive, and the placebo that separates a persona from the mere presence of an extra sentence. Three families are used. The first assigns no persona at all, and is the neutral condition behind every baseline number in this supplement. The second is a risk-aware family with six ordered levels, R1 to R6, whose wording is adapted from the six-choice Eckel–Grossman gamble scale so that R1 is the most risk-averse framing and R6 the least. The third is a social family that frames the rival either as cooperative or as adversarial. The neutral placebo adds a sentence of matched length that carries no risk content and no social content, so a difference between the placebo and no persona at all is an effect of prompt length rather than of persona. All of these are prompt conditions written into the instruction the model receives. None is a trait measured on the agent, and nothing here is elicited from the model, which is the distinction the rest of this section turns on.

to different checkpoints. Of the seven tested checkpoints, five have such a sweep; neither Gemini Flash-Lite checkpoint does. In logistic fits that additionally control for the mechanism's actual private-risk treatment, the coefficients on the agent's own assigned risk label remain large for the three checkpoints where that controlled fit has been run, so the persona effect is not simply standing in for the mechanical risk treatment.

This contrast does not show that LLM agents possess a latent risk preference comparable to the human elicitation: the human variable is a measured pre-game disposition that turns out to be nearly unrelated to in-game behaviour, whereas the LLM variable is an explicit instruction embedded in the prompt. The more defensible reading is that an assigned risk persona functions as a high-salience behavioural instruction for these models, not as an analogue of a

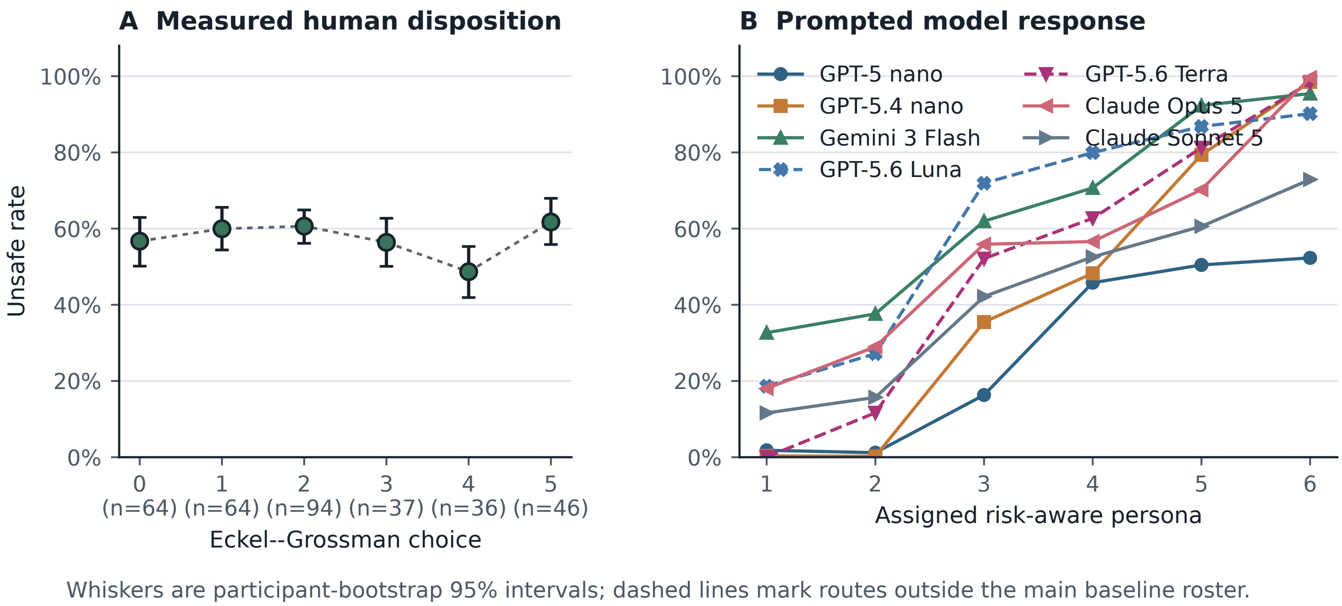


Figure 4: A measured human disposition against an assigned model label. Left: mean UNSAFE rate by elicited Eckel–Grossman gamble choice, with 95% confidence intervals and participants per bin; the dotted line is the trend across the six group means. Right: mean UNSAFE rate by assigned risk-persona level, for every model with a six-level sweep; GPT-5.6 Luna and Terra (dashed) are the two routes outside the main baseline roster. The shared six-step axis is visual alignment only; the left is a disposition measured once before play, the right an instruction re-assigned every race.

Table 9: LLM risk-persona effect: change in UNSAFE rate from the lowest (R1) to the highest (R6) assigned risk label, and the per-level OLS slope, for every checkpoint with a six-level risk-label sweep ($n = 360$ players per level, $n = 330$ for Gemini-3-Flash). The persona coefficient (a logistic fit that additionally controls for the mechanism's actual private-risk treatment) has been run only for the first three; "not estimated" marks checkpoints for which only the descriptive change and slope are available. Starred rows sit outside the seven-checkpoint roster.

Checkpoint	Δ UNSAFE (pp)	Per-level slope	Coefficient
GPT-5-nano	50.5	0.123	0.93
GPT-5.4-nano	98.2	0.212	1.47
Gemini-3-Flash	62.8	0.139	2.03
Claude Opus 5	81.6	0.152	not estimated
Claude Sonnet 5	61.3	0.129	not estimated
GPT-5.6 Luna*	71.5	0.156	not estimated
GPT-5.6 Terra*	98.3	0.203	not estimated

stable human risk disposition. Because the LLM estimates come from a separate risk-matrix experiment, available for five of the seven tested checkpoints and with the private-risk-controlled logistic fit run for only three of those five, this comparison remains exploratory and cross-experimental rather than a like-for-like measurement.

Table 10: Race-grouped population-identity diagnostic on 760 complete first-five-round trajectories. The eight populations are imbalanced (340 human and 60 per checkpoint), so balanced accuracy and macro-F1 are primary. The null interval is the 2.5–97.5 percentile range over grouped label permutations.

Metric	Observed	Permutation null mean	Null 95% interval
Raw accuracy	0.292 ± 0.053	0.086	0.039–0.178
Balanced accuracy	**0.423 ± 0.043**	0.124	0.070–0.182
Macro-F1	**0.252 ± 0.047**	0.063	0.025–0.112

A.11 Population identity in first-five-round trajectories

Table 10 is the diagnostic the main paper summarises: it asks how well a simple classifier can name the population that produced a trajectory, using the same 15 raw features as the trajectory embedding of §A.15. We fit a shallow decision tree with class-balanced loss under stratified five-fold grouped cross-validation. The two seats of an LLM race remain in the same fold; human observations are grouped by participant. Since the pooled classes are unequal, balanced accuracy and macro-F1 are the primary measures. A 200-repetition permutation null preserves the observed class counts and the same folds.

The balanced accuracy exceeds the class-balanced chance level of 1/8 and the grouped permutation null ($p < 0.001$); raw accuracy is shown only for completeness because it is affected by the large human class. This result supports observable differences in the first-five-round trajectories, while the overlap and the shallow classifier do not support a claim that a latent strategy or model identity is cleanly recoverable. The machine-readable result and source hashes are stored in `results/cross_model_pilot_synthesis/data/`.

A.12 Predictive structure of UNSAFE choices

The preceding section shows that populations produce different action patterns. We now ask which recorded parts of the game best predict their next UNSAFE choice. This is a study of association, not cause or internal reasoning. For each population, we fit a random forest, a model that combines many decision trees. Table 11 reports the result and Figure 5 shows the same shares as a heatmap. Predictive scores use five-fold grouped cross-validation, with all decisions from one participant or race kept in one fold. The forest predicts decisions after round one from five values known before the choice: the player's previous action, the opponent's previous action, the progress gap, the assigned maximum private risk, and the round number. TreeSHAP then measures how much each value contributes to the fitted model's predictions. A larger SHAP share means that the predictor relied more on that feature. It does not mean that the feature caused the decision.

For human participants, the opponent's previous action accounts for 56% of total SHAP magnitude, far more than any other feature, while the player's own previous action contributes only 5%: a reciprocity-dominated profile that independently recovers the pattern found in the human logistic analysis. GPT-5-nano is close to the opposite profile: relative race position (progress gap) accounts for 41% of its predictive importance and opponent history for only 7%, and its apparently strong out-of-fold AUC of 0.80 is misleading on its own, since balanced accuracy is exactly 0.50: the checkpoint plays so close to the SAFE floor that the classifier effectively fails to recover the rare UNSAFE class. Gemini-3-Flash and Gemini-3.1-Flash-Lite are primarily risk-treatment-driven (33% and 37% of SHAP magnitude), consistent with their strong, monotone descriptive risk-response curves. Claude Sonnet 5 is the only model whose profile matches the human ordering outright, with the opponent's previous action largest at 48%, below the human 56%; Gemini-3.5-Flash-Lite is the next closest (33%), though its progress gap remains more influential than in the human model. GPT-5.4-nano has no comparably dominant predictor: importance is spread across progress gap, risk, round, and its own previous action, with opponent history at 10% and the lowest above-chance predictive fit in the table (AUC 0.54, balanced accuracy 0.52), so its behaviour is comparatively difficult to reconstruct from these five mechanical state variables rather than clearly governed by any one signal.

Claude Opus 5 shows why these shares must be read against the design that produced them. Its 42% opponent share reads as strong reciprocity and is not: on the neutral lane it plays UNSAFE on essentially every low-risk decision and essentially none at high risk, so the risk treatment alone separates its choices, and since both seats behave that way the opponent's previous action is a proxy for the risk level rather than an independent signal. The forest then splits importance between two variables encoding the same event, and the perfect AUC describes a step function, not a better fit. A feature-importance share is interpretable only where the feature keeps variation independent of the others, which is what a near-deterministic policy removes.

Predictive performance varies substantially across populations (AUC 0.54 to 1.00), so SHAP shares should not be read as directly comparable measures of decision quality; floor and ceiling behaviour mechanically limits minority-class prediction for several models. The defensible conclusion is narrower than "LLMs use different features": the same observable state variables organise predictive information differently across populations, and matching a human aggregate action rate does not imply matching the human reciprocity-dominated association profile.

A.13 Two-player checkpoint descriptive baseline (full scope)

The five-checkpoint two-player descriptive baseline referenced throughout the two-player strategic diversity results in the main paper and the observable drivers of unsafe play in the main paper completed 150 races and 2,790 decisions, crossing five checkpoints (GPT-5-nano, GPT-5.4-nano, Gemini-3-Flash-preview, Gemini-3.1-Flash-Lite, Gemini-3.5-Flash-Lite) with the mechanism's three assigned private-risk levels ($0.10, 0.60, 0.90$; Eq. (2)). At the level of descriptive curves rather than the pooled aggregate rates already reported in the main text, GPT-5-nano's UNSAFE rate was low and nearly flat across the three risk levels; GPT-5.4-nano's response to risk level was non-monotone; and the three Gemini checkpoints' UNSAFE rate declined as the assigned maximum risk increased, the opposite direction from the two OpenAI checkpoints. Because provider routes, protocol signatures, and pilot sample sizes differ across checkpoints, these five curves demonstrate qualitative heterogeneity in how each checkpoint responds to the risk manipulation; we do not pool them into a single model-family estimate, and the formal test of that heterogeneity is the checkpoint-by-risk interaction reported in the two-player strategic diversity results in the main paper ($\chi^2(10) = 354.7$, with a convergence warning, so only the conservative $p < 10^{-3}$ threshold is used in the main text). Table 12 gives the per-checkpoint, per-risk-level rates underlying the qualitative curve descriptions above.

A.14 N-player self-play scope ($N = 3, 4, 5$)

The exploratory multiplayer position analysis reports position effects for GPT-5-nano and GPT-5.4-nano at fixed N; this subsection gives the broader group-size scope. Matched OpenAI self-play baselines cover $N \in \{3, 4, 5\}$ for both checkpoints: 180 pilot races, 720 player trajectories, and 6,120 decisions in total, pooled across the mechanism's three risk levels. Table 13 reports the aggregate UNSAFE rate by checkpoint and group size.

Neither checkpoint's response to group size is monotone in N, and the association is checkpoint-specific: GPT-5-nano's rate roughly doubles from $N = 3$ to $N = 4$ before partially receding at $N = 5$, while GPT-5.4-nano stays within a narrow 83–87% band throughout. We do not read either pattern as an isolated group-size effect, for two reasons. First, changing N also changes the stage-payoff rule itself (Eq. (3)–(4) use the joint count k^t of SAFE-choosing companies rather than a single opponent's move), the number of opponents, and prompt length, so N is confounded with the representation of the game as well as with its size. Second,

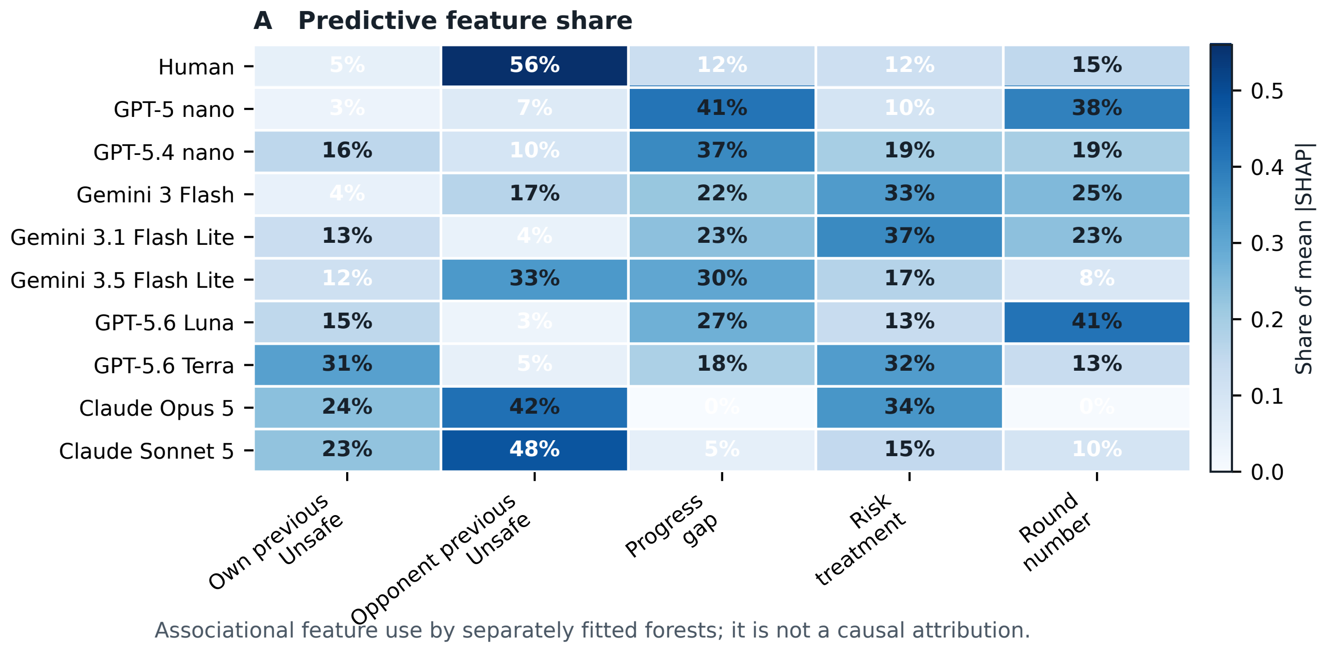


Figure 5: Population-specific predictive structure for round-$t \geq 2$ UNSAFE choices. Each row reports the share of mean absolute SHAP value assigned to the same five pre-decision features by a random forest fitted separately to that population. SHAP magnitudes describe how the fitted classifier uses observed features; they are neither causal effects nor evidence of the agents' internal reasoning.

Table 11: Population-specific random-forest prediction of UNSAFE play. Feature columns report shares of total mean absolute SHAP magnitude (rows sum to 100%, up to rounding); AUC and balanced accuracy are mean out-of-fold values from the population-specific race/participant-grouped five-fold split.

Population	Own previous	Opponent previous	Progress gap	Risk	Round	OOF AUC	Balanced accuracy
Human	5%	**56%**	12%	12%	15%	0.63	0.58
GPT-5-nano	3%	7%	**41%**	10%	38%	0.80	0.50
GPT-5.4-nano	16%	10%	**37%**	19%	19%	0.54	0.52
Gemini-3-Flash	4%	17%	22%	**33%**	25%	0.92	0.77
Gemini-3.1-Flash-Lite	13%	4%	23%	**37%**	23%	0.95	0.75
Gemini-3.5-Flash-Lite	12%	**33%**	30%	17%	8%	0.83	0.71
Claude Opus 5	24%	42%†	0%	34%	0%	1.00	1.00
Claude Sonnet 5	23%	**48%**	5%	15%	10%	0.97	0.93

†Collinear with the risk treatment, not a reciprocity estimate; see the discussion below.

Table 12: Player-level UNSAFE rate by checkpoint and assigned maximum-private-risk level, five-checkpoint two-player descriptive baseline (n = 20 players per checkpoint–risk cell, ten races each).

Checkpoint	Risk 0.10	Risk 0.60	Risk 0.90
GPT-5 nano	12.3%	15.1%	14.5%
GPT-5.4 nano	57.7%	49.6%	56.7%
Gemini 3 Flash	100.0%	72.3%	53.9%
Gemini 3.1 Flash Lite	100.0%	80.1%	69.9%
Gemini 3.5 Flash Lite	83.8%	70.8%	62.6%

Table 13: Aggregate UNSAFE rate by group size, OpenAI self-play, pooled across the three assigned private-risk levels. Ten independent races per model–risk–N cell.

Checkpoint	$N = 3$	$N = 4$	$N = 5$
GPT-5-nano	16.2%	26.8%	21.9%
GPT-5.4-nano	83.7%	87.0%	84.0%

the pilots use only ten independent races per model–risk–N cell, which is a small base for the per-cell rates entering Table 13 and leaves individual cells vulnerable to the same kind of small-n noise flagged for specific rank × risk-band cells in the exploratory position analysis. Table 14 disaggregates Table 13 by assigned risk level and reports a 95% percentile bootstrap interval over the ten independent races in each cell. The decision count is retained only as a descriptive row count.

Table 14: UNSAFE rate by checkpoint, group size, and assigned maximum-private-risk level, OpenAI N-player self-play. Intervals are 95% percentile bootstrap intervals over race clusters, and n_{race} is the independent-unit count.

N	Checkpoint	Risk	n_{race}	UNSAFE rate (95% CI)
3	GPT-5 nano	0.10	10	17.4% (14.5–20.0%)
3	GPT-5 nano	0.60	10	18.1% (13.3–22.4%)
3	GPT-5 nano	0.90	10	11.3% (6.1–16.9%)
3	GPT-5.4 nano	0.10	10	85.4% (82.5–88.5%)
3	GPT-5.4 nano	0.60	10	83.7% (78.5–88.7%)
3	GPT-5.4 nano	0.90	10	83.3% (80.0–86.9%)
4	GPT-5 nano	0.10	10	19.2% (14.5–23.8%)
4	GPT-5 nano	0.60	10	27.8% (21.6–33.3%)
4	GPT-5 nano	0.90	10	24.1% (15.5–32.1%)
4	GPT-5.4 nano	0.10	10	88.1% (85.9–90.3%)
4	GPT-5.4 nano	0.60	10	88.4% (86.2–90.6%)
4	GPT-5.4 nano	0.90	10	83.8% (80.3–87.5%)
5	GPT-5 nano	0.10	10	18.9% (14.7–22.4%)
5	GPT-5 nano	0.60	10	20.0% (16.1–23.6%)
5	GPT-5 nano	0.90	10	22.7% (18.7–26.9%)
5	GPT-5.4 nano	0.10	10	80.0% (75.4–84.9%)
5	GPT-5.4 nano	0.60	10	87.1% (84.5–89.6%)
5	GPT-5.4 nano	0.90	10	85.1% (82.5–88.0%)

A.15 Trajectory embedding recoloured by outcome

The main paper reports that the embedding of raw five-round trajectories separates almost entirely by how much UNSAFE play a trajectory carries. Both colourings of that embedding now live here. Figure 6 colours it by population, and Figure 7 is the check: the same coordinates, with only the colouring changed, so any structure visible in the second is structure the embedding already had. Reading them in that order is the point. The population panels look like eight different regions; recolouring the identical points by each player's own UNSAFE rate splits the embedding into a SAFE region, an UNSAFE region and a thin mixed band, with very few points on the wrong side. The apparent separation of populations is therefore largely a separation of action rates, which is why the main paper rests no conclusion on either picture and uses the grouped classifier of Table 10 instead.

A.16 Mean round-by-round profile per population

The embedding in the main paper shows that human trajectories spread more widely than any single checkpoint's. Figure 8 is the same comparison read one feature at a time: population means over the fourteen features every trajectory shares, with action probabilities and absolute progress gaps separated so that neither scale obscures the other.

Table 15: UNSAFE rate by persona band and relative position, GPT-5-nano. n is the number of decisions in the cell. The interval is a 95% Wilson score interval computed at the level of the decision. Decisions within a race are not independent, and the race is this paper's independent unit, so the interval is anti-conservative: it is narrower than a race-clustered interval would be. These cells are descriptive only and support no inference.

Persona band	Position	n	UNSAFE rate	95% CI
Baseline	Leader	1460	19.7%	17.7–21.8%
Baseline	Middle	1065	30.4%	27.7–33.3%
Baseline	Trailer	175	33.1%	26.6–40.4%
Low (R1–R2)	Leader	1094	5.9%	4.6–7.4%
Low (R1–R2)	Middle	198	14.6%	10.4–20.2%
Low (R1–R2)	Trailer	58	12.1%	6.0–22.9%
Mid (R3–R4)	Leader	649	40.7%	37.0–44.5%
Mid (R3–R4)	Middle	451	38.1%	33.8–42.7%
Mid (R3–R4)	Trailer	250	32.8%	27.3–38.8%
High (R5–R6)	Leader	702	63.2%	59.6–66.7%
High (R5–R6)	Middle	339	54.3%	49.0–59.5%
High (R5–R6)	Trailer	309	50.2%	44.6–55.7%

A.17 Player-level UNSAFE rate by population

Figure 9 is the aggregate level of the same comparison: one number per population, the mean share of a player's first five decisions that were UNSAFE. It is the plainest form of the point that a matching aggregate rate does not imply matching behaviour, because a population can reach a middling rate either by mixing or by every player behaving the same way.

A.18 UNSAFE rate by persona band and relative position

The main paper does not report these cells. Tables 15 and 16 give them in full for the two checkpoints that have a persona sweep and an observed position, so a single rate and its interval can be checked directly. Relative position is observed rather than assigned, so these are associations within a persona band, not position effects. One further caveat governs both tables and is stated in each caption as well: the intervals are Wilson score intervals computed over *decisions*, whereas this paper's method statement makes the race the independent unit. Decisions taken inside one race are not independent of each other, so a decision-level interval is narrower than the race-level uncertainty warrants. The per-race source for these two specific tables could not be located in the analysis record, so we neither recompute the intervals nor delete the cells: they are printed as descriptive breakdowns, and no inference rests on them.

A.19 Cluster-count sensitivity and stability of the human archetypes

The behavioural archetypes used in this supplement and in the main paper are k-means groups: the 341 human participants are described by five standardised summary numbers each, and k-means partitions them into k groups so that each participant is near the average of its own group. The number of groups is a choice, not a

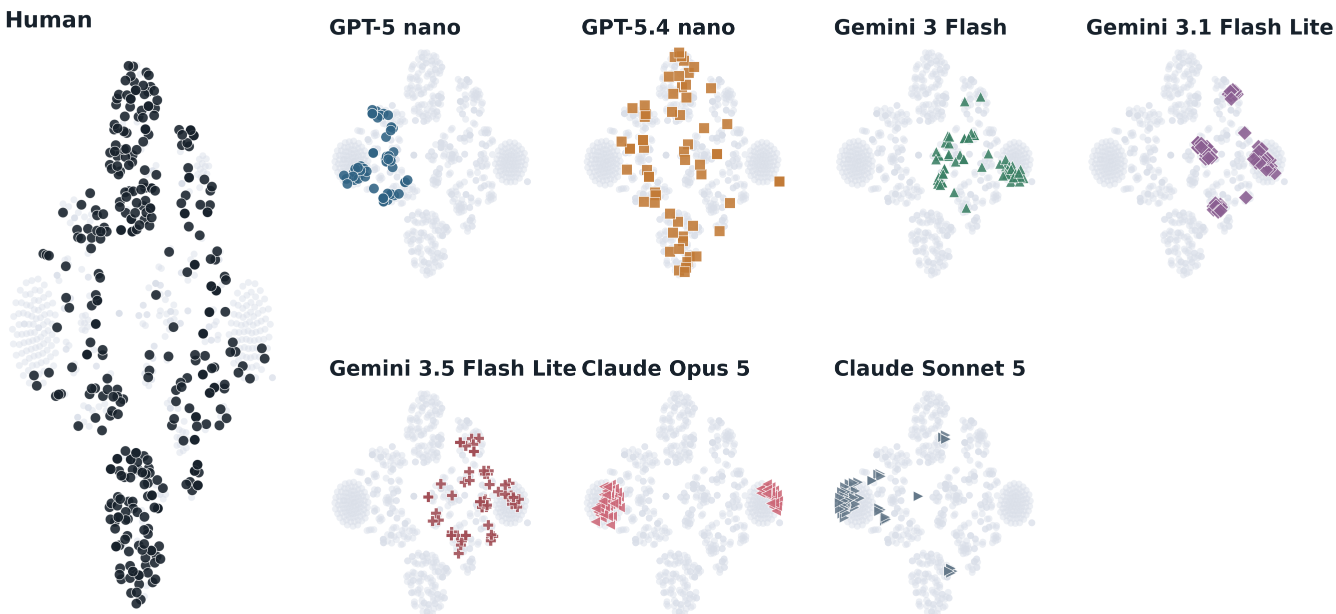


Figure 6: The population-coloured embedding, shown here so that Figure 7 can be compared against it directly. A t-SNE embedding places each of the 760 complete first-five-round trajectories at a point in two dimensions, arranged so that trajectories with similar action-and-position histories fall near each other; the input is the pooled 15-dimensional first-five-round action-and-position space, the same input as the population-identity classifier below. One panel is drawn per population, human first. The axes have no units and no direction: only relative closeness carries meaning, and distances between well-separated clusters are not interpretable. The main paper rests no conclusion on this figure, and the reason is Figure 7: recolouring these identical coordinates by each player's own UNSAFE rate separates the embedding almost entirely, so what looks like a separation of populations here is largely a separation of action rates. The quantitative statement about population identity is the grouped classifier of Table 10, not this picture.

Table 16: UNSAFE rate by persona band and relative position, GPT-5.4-nano. n is the number of decisions in the cell and cells marked † have $n < 20$. The interval is a 95% Wilson score interval computed at the level of the decision. Decisions within a race are not independent, and the race is this paper's independent unit, so the interval is anti-conservative: it is narrower than a race-clustered interval would be. These cells are descriptive only and support no inference.

Persona band	Position	n	UNSAFE rate	95% CI
Baseline	Leader	1649	83.6%	81.8–85.3%
Baseline	Middle	703	84.8%	81.9–87.2%
Baseline	Trailer	348	86.5%	82.5–89.7%
Low (R1–R2)	Leader	1073	3.3%	2.4–4.5%
Low (R1–R2)	Middle	268	1.5%	0.6–3.8%
Low (R1–R2)	Trailer	9†	0.0%	0.0–29.9%
Mid (R3–R4)	Leader	1164	95.7%	94.4–96.7%
Mid (R3–R4)	Middle	70	88.6%	79.0–94.1%
Mid (R3–R4)	Trailer	116	91.4%	84.9–95.3%
High (R5–R6)	Leader	1328	99.3%	98.7–99.6%
High (R5–R6)	Middle	4†	100.0%	51.0–100.0%
High (R5–R6)	Trailer	18†	100.0%	82.4–100.0%

measurement, so this subsection reports what changes when that choice changes. The five descriptors are the overall UNSAFE rate, reciprocity, position sensitivity, own-action autocorrelation, and whether the first round was UNSAFE. They are standardised on the human sample alone, and every other population is later projected into that same human space rather than re-standardised.

Table 17 gives the diagnostics for $k = 2$ to 6. Three of them describe how tight and separated the groups are: the silhouette coefficient, which is higher when a participant is closer to its own group than to the nearest other one; the Calinski–Harabasz index, higher when between-group spread is large relative to within-group spread; and the Davies–Bouldin index, which is *lower* when groups are better separated. The fourth, inertia, is the total squared distance to group centres and falls mechanically as k grows, so it is reported for completeness rather than as evidence. The choice of $k = 4$ used elsewhere is the best cell on two of the three interpretable indices, with the highest silhouette (0.316, against 0.278 at $k = 2$ and 0.241 at $k = 5$) and the lowest Davies–Bouldin (1.218).

Stability is only moderate, and we say so rather than presenting the groups as recovered types. The bootstrap adjusted Rand index falls from 0.961 at $k = 2$, where the split is nearly always the same one, to 0.717 at $k = 3$, 0.626 at $k = 4$, 0.458 at $k = 5$, and 0.518 at $k = 6$. At $k = 4$ the interval runs from 0.39 to 0.98, which is wide. The groups are therefore useful descriptions of where the

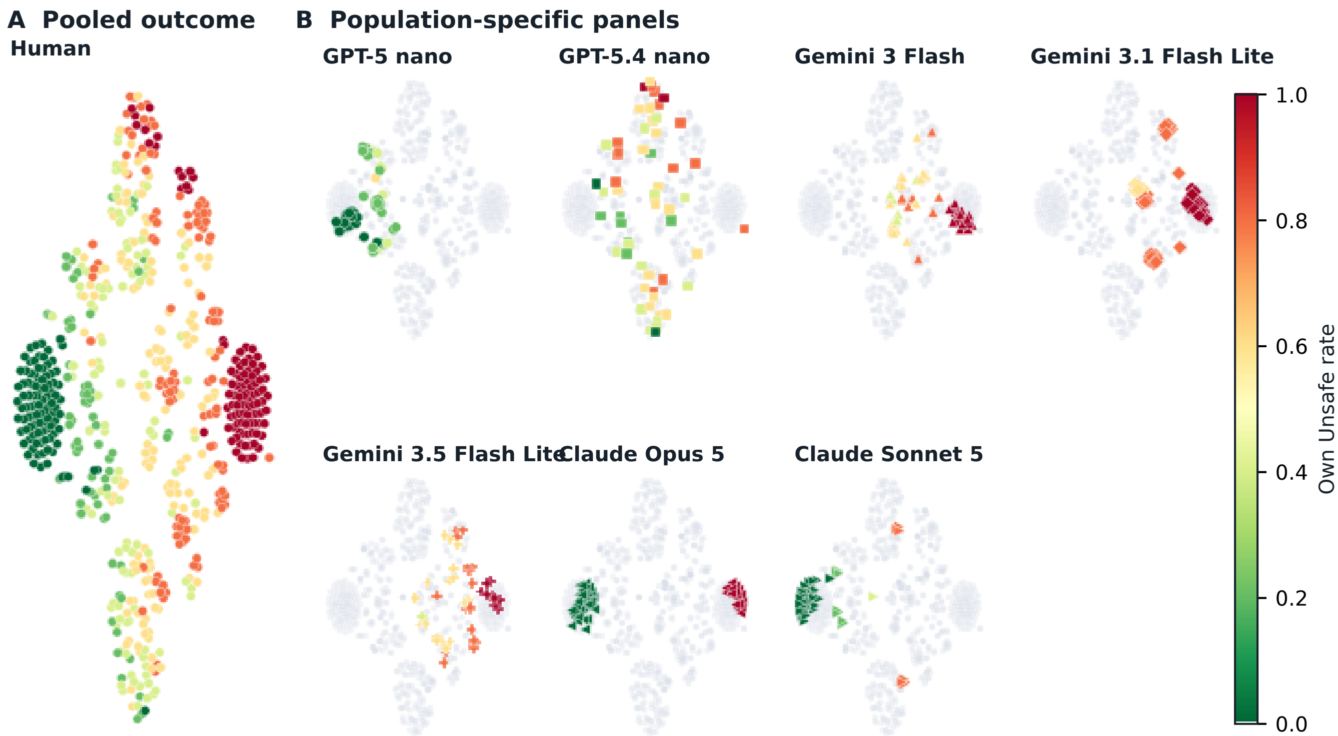


Figure 7: The same t-SNE coordinates as the population-identity embedding in the main paper, recoloured by each player's own mean UNSAFE rate over rounds 1–5 (green = SAFE, red = UNSAFE) instead of population identity. Left: all 760 trajectories pooled. Right: the same colouring broken out per population.

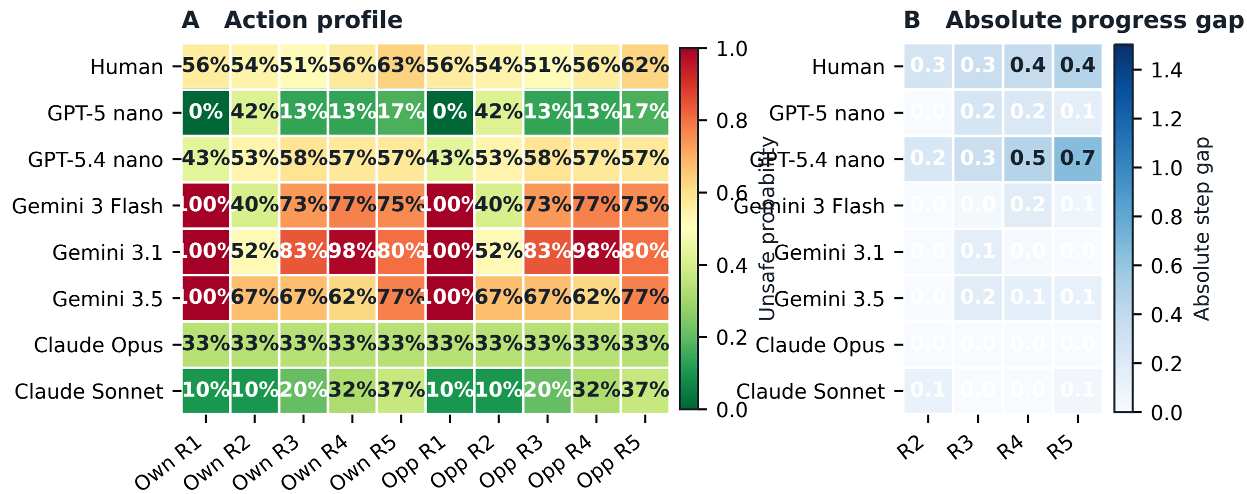
Table 17: Cluster-count sweep for the human behavioural archetypes, $k = 2$ to 6, on the same matrix of 341 human participants by five standardised descriptors, seed 0. Higher is better for silhouette and Calinski–Harabasz, lower is better for Davies–Bouldin; inertia falls with k by construction. "Group sizes" is the number of participants per group. The adjusted Rand index (ARI) measures how much a partition refitted on a resampled sample agrees with the partition fitted on the full sample: 1 is identical, 0 is chance agreement. It is computed over 500 bootstrap resamples of participants, and the interval is the 2.5 to 97.5 percentile range across those resamples.

k	Silhouette	Calinski Harabasz	Davies Bouldin	Inertia	Group sizes	Smallest group n	ARI mean	ARI 95% interval
2	0.278	110.9	1.578	1280.9	187 / 154	154	0.961	0.76–1.00
3	0.267	91.7	1.542	1102.1	107 / 63 / 171	63	0.717	0.37–0.98
4	0.316	83.5	1.218	975.3	19 / 170 / 133 / 19	19	0.626	0.39–0.98
5	0.241	78.2	1.363	880.2	20 / 57 / 118 / 87 / 59	20	0.458	0.30–0.69
6	0.239	79.0	1.405	780.2	42 / 78 / 56 / 55 / 91 / 19	19	0.518	0.34–0.79

human sample sits in this five-number space, not evidence that four discrete human strategies exist.

The conclusion the main paper draws from these groups does not depend on k. That conclusion is about coverage: the human sample spreads across the whole set of archetypes while almost every single checkpoint concentrates on a few. Table 18 counts, for each population and each k, how many of the k archetypes that population actually occupies, where a population counts as occupying an archetype when at least 5% of its entities project to it. The human row occupies all k archetypes at every k from two to six. No checkpoint matches that. The closest is the exception the main paper names, GPT-5.4 nano, which occupies all k archetypes at $k = 2, 3, 5$ and 6 and three of four at $k = 4$. Every other checkpoint falls short from $k = 4$ upward, and GPT-5 nano is the extreme case, occupying exactly one archetype at every k. The 5% threshold is a reporting choice, and the artifact records the counts at a 1% threshold and at strictly positive occupancy alongside it so the claim can be read at any of the three.

One property of these descriptors is ours to disclose rather than a reviewer's to discover. Two of the five are undefined for some



Each row is a population mean over complete first-five trajectories; absolute gap is measured entering rounds 2--5.

Figure 8: Round-by-round profile for each population, shown as two source-backed heatmaps. Panel A reports the probability of an UNSAFE action for each player's own and opponent action at rounds 1–5. Panel B reports the mean absolute progress gap entering rounds 2–5, so it measures how far apart the two players are after pooling focal-player roles. The eight rows are the human reference and the seven baseline model populations.

Table 18: Archetype occupancy by population and cluster count. Each cell counts how many of the k human-fitted archetypes that population occupies, where occupying means that at least 5% of the population's entities project nearest to that archetype. n is the number of entities projected: participants for the human row, no-persona player-races for each checkpoint. A row that reaches k at every k covers the full human-fitted space; the human reference does so at every k and no checkpoint does.

Population	n	k=2	k=3	k=4	k=5	k=6
Human (reference)	341	2	3	4	5	6
GPT-5 nano	120	1	1	1	1	1
GPT-5.4 nano	120	2	3	3	5	6
Gemini 3 Flash	120	1	1	2	3	3
Gemini 3.1 Flash Lite	60	1	1	2	2	3
Gemini 3.5 Flash Lite	60	1	1	2	3	3
GPT-5.6 Luna	120	2	1	2	2	3
GPT-5.6 Terra	120	2	2	3	3	4
Claude Opus 5	120	2	2	2	2	2
Claude Sonnet 5	120	2	3	2	3	4

participants, and the clustering fills the gap with the sample mean instead of dropping the participant, exactly as the published clustering does. Position sensitivity, which contrasts a participant's behaviour when ahead against when behind, is undefined for 268 of the 341 participants, because those participants were never observed both ahead and behind; it is mean-imputed for all 268, that is for 79% of the sample. Reciprocity and own-action autocorrelation are each undefined and mean-imputed for 38 participants. The overall UNSAFE rate and the first-round action are complete for all 341. A mean-imputed cell carries no information about the participant it is attached to, so position sensitivity does most of its work on under a quarter of the sample, and the archetype signatures should be read with that in mind. No participant is excluded at any step: all 341 receive a cluster label.

A.20 Trajectory-diversity statistics

The main paper reports that human first-five-round trajectories are more varied than any single checkpoint's. Table 19 gives the numbers behind that, one row per population per risk level, so a single value can be checked against its population, its risk condition, and the number of independent units it rests on.

A trajectory here is the pair of actions taken by the two players over the first five rounds, that is ten SAFE or UNSAFE bits. Three statistics describe how varied a set of such trajectories is. Hill $q = 0$ counts how many *distinct* trajectories appear, so it treats a trajectory seen once and a trajectory seen forty times alike. Hill $q = 1$ is the effective number of trajectories, the exponential of Shannon entropy, which discounts trajectories that are rare: it equals $q = 0$ when every trajectory is equally common and falls below it when a few dominate. Mean pairwise Hamming distance is the average number of the ten bits on which two trajectories drawn from the set differ, divided by ten, so 0 means every trajectory in the set is identical and 0.5 means two trajectories differ on half their bits on average.

Sample sizes differ across populations, and both counting statistics rise with sample size, so every cell is compared at the same size. The point estimate is the mean over draws of 20 trajectories

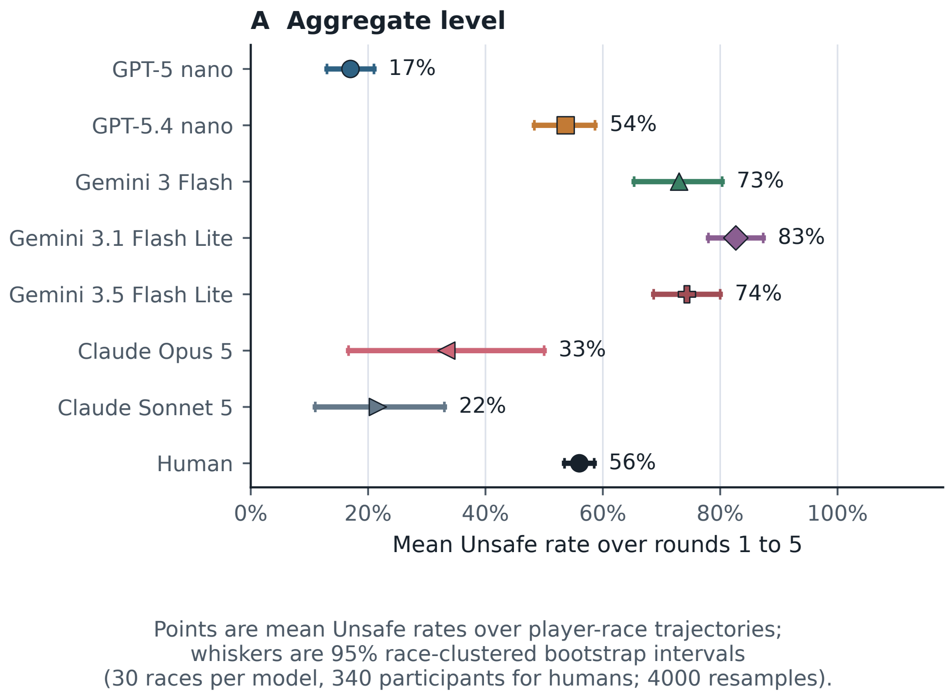


Figure 9: Player-level mean UNSAFE rate over rounds 1 to 5, one row per population. The point is the unweighted mean over player-race trajectories. The whisker is a 95% percentile bootstrap interval that resamples *clusters*, not single trajectories, because the two seats of a race are not independent of each other: they share the sampled stopping horizon and the setback draws. The cluster is therefore the race for every checkpoint row, 30 races each, and the participant for the human row, 340 participants; the interval uses 4,000 resamples. The point estimates are unchanged from a trajectory-level calculation; only the interval changed, and it supersedes an earlier interval that resampled player-race trajectories and so treated the two seats of a race as independent.

taken without replacement from the cell, or the exact cell value where the cell already holds exactly 20. The interval is a two-stage cluster bootstrap: clusters are resampled with replacement first, and only then is the draw rarefied to 20 inside each resample, over 2,000 resamples. The cluster is the independent unit of the design: a participant for the human rows, one per trajectory, and a race for the checkpoint rows, because the two seats of a race share the same sampled horizon and setback draws and so are not independent of each other. The n_{cluster} column is that count.

The ordering in Table 19 is consistent across all three risk levels and all three statistics. The human rows sit near the ceiling of the comparison: 19.2, 19.7, and 19.5 distinct trajectories out of a possible 20, with mean pairwise Hamming distance between 0.467 and 0.497, close to the 0.5 that two unrelated trajectories would give. Only one checkpoint reaches that: GPT-5.4 nano returns 19.0 distinct trajectories at every risk level, with Hamming between 0.478 and 0.506. The rest are far below. Claude Opus 5 returns exactly one distinct trajectory at all three risk levels, with Hamming 0, meaning every one of its player-races produced the identical ten-action sequence. Gemini 3 Flash and Gemini 3.1 Flash Lite collapse the same way at risk 0.1. Claude Sonnet 5 falls from 5.0 distinct trajectories at risk 0.1 to 3.0 at risk 0.9, with Hamming 0.020 at the latter. This is what the main paper's diversity claim rests on: a checkpoint can match a human aggregate action rate while producing a small handful of distinct action sequences, or only one.

A.21 Behavioural-archetype coverage by population

The main paper reports that human trajectories spread more evenly across the archetypes than most single checkpoints do, with GPT-5.4-nano the exception. Table 20 is the per-population breakdown behind that statement at the four archetypes used throughout, so a single share can be checked against its population and its archetype; Table 18 is the same comparison read as a count, at every cluster count from two to six.

A.22 Matched group-size comparison

The three- to five-player pilots reported elsewhere in this supplement have no two-player arm, so a difference between group sizes could not be told apart from a difference between two separately built lanes. This subsection reports a matched comparison that removes that particular ambiguity.

Every group size, including two, is run through the same N-player engine, the same prompt template, the same parser and one protocol, `ai-race-nplayer-matched-hosted-confirmatory-v1`,

Table 19: Sample-size-matched diversity of paired first-five-round trajectories, by maximum-private-risk level and population. A trajectory is the two players' ten actions over rounds 1 to 5. Hill q=0 is the number of distinct trajectories, Hill q=1 the effective number after discounting rare ones, and Hamming the mean fraction of the ten bits on which two trajectories differ. n_{src} is the trajectories available in the cell and n_{cluster} the independent units it rests on: participants for the human rows, races for the checkpoint rows. Every cell is compared at 20 trajectories. Intervals are 95% two-stage cluster bootstrap intervals: clusters resampled with replacement, then rarefied to 20 within each resample, 2,000 resamples. Because the point estimate is the observed cell value while the interval is computed on resampled clusters, which lose distinct trajectories to duplication, an interval can sit below its point estimate; the interval bounds resampling variability and is not a confidence statement about the observed count.

Risk	Population	n_{src}	Hill q=0 (95% CI)	Hill q=1 (95% CI)	Hamming (95% CI)	n_{cluster}
0.1	Human	98	19.2 (15.0–20.0)	18.9 (13.2–20.0)	0.467 (0.407–0.504)	98
	GPT-5 nano	20	13.0 (5.0–12.0)	10.2 (3.4–10.7)	0.275 (0.163–0.339)	10
	GPT-5.4 nano	20	19.0 (9.0–16.0)	18.7 (7.3–15.2)	0.478 (0.405–0.505)	10
	Gemini 3 Flash	20	1.0 (1.0–1.0)	1.0 (1.0–1.0)	0.000 (0.000–0.000)	10
	Gemini 3.1 Flash Lite	20	1.0 (1.0–1.0)	1.0 (1.0–1.0)	0.000 (0.000–0.000)	10
	Gemini 3.5 Flash Lite	20	5.0 (3.0–5.0)	3.6 (1.9–4.8)	0.166 (0.076–0.227)	10
	Claude Opus 5	20	1.0 (1.0–1.0)	1.0 (1.0–1.0)	0.000 (0.000–0.000)	10
	Claude Sonnet 5	20	5.0 (3.0–5.0)	4.5 (2.6–5.0)	0.354 (0.246–0.413)	10
0.6	Human	104	19.7 (15.0–20.0)	19.7 (14.1–20.0)	0.495 (0.455–0.513)	104
	GPT-5 nano	20	8.0 (5.0–8.0)	7.2 (3.4–7.7)	0.244 (0.154–0.301)	10
	GPT-5.4 nano	20	19.0 (9.0–16.0)	18.7 (7.2–15.2)	0.506 (0.402–0.520)	10
	Gemini 3 Flash	20	11.0 (5.0–11.0)	9.3 (4.4–10.2)	0.312 (0.208–0.371)	10
	Gemini 3.1 Flash Lite	20	4.0 (2.0–4.0)	2.8 (2.0–3.7)	0.229 (0.211–0.255)	10
	Gemini 3.5 Flash Lite	20	12.0 (6.0–12.0)	11.5 (5.5–10.9)	0.364 (0.267–0.397)	10
	Claude Opus 5	20	1.0 (1.0–1.0)	1.0 (1.0–1.0)	0.000 (0.000–0.000)	10
	Claude Sonnet 5	20	5.0 (1.0–5.0)	2.7 (1.0–4.0)	0.074 (0.000–0.133)	10
0.9	Human	138	19.5 (16.0–20.0)	19.3 (14.8–20.0)	0.497 (0.457–0.516)	138
	GPT-5 nano	20	12.0 (6.0–12.0)	10.0 (3.9–10.7)	0.232 (0.134–0.304)	10
	GPT-5.4 nano	20	19.0 (9.0–16.0)	18.7 (7.3–15.2)	0.486 (0.394–0.504)	10
	Gemini 3 Flash	20	11.0 (6.0–11.0)	10.7 (5.5–10.2)	0.311 (0.262–0.314)	10
	Gemini 3.1 Flash Lite	20	3.0 (2.0–3.0)	2.9 (2.0–3.0)	0.101 (0.079–0.105)	10
	Gemini 3.5 Flash Lite	20	12.0 (6.0–12.0)	11.5 (5.2–10.9)	0.401 (0.319–0.417)	10
	Claude Opus 5	20	1.0 (1.0–1.0)	1.0 (1.0–1.0)	0.000 (0.000–0.000)	10
	Claude Sonnet 5	20	3.0 (1.0–3.0)	1.5 (1.0–2.3)	0.020 (0.000–0.054)	10

Table 20: Behavioural-archetype coverage: share of each population's neutral-lane player-races assigned to each human-fit archetype. Human row is the reference clustering (n=341); each LLM row projects that checkpoint's own no-persona player-races (n=60 to 120) into the same space. This table draws on the full nine-checkpoint cross-model pilot and therefore includes GPT-5.6 Luna and Terra, which are not part of the seven-checkpoint trajectory comparison in the main paper.

Population	Cautious	Aggr./recip.	Catch-up	Persister
Human (reference)	39.0%	49.9%	5.6%	5.6%
GPT-5-nano	99.2%	0%	0%	0.8%
GPT-5.4-nano	51.7%	33.3%	3.3%	11.7%
Gemini-3-Flash	1.7%	78.3%	20.0%	0%
Gemini-3.1-Flash-Lite	0%	83.3%	16.7%	0%
Gemini-3.5-Flash-Lite	0%	76.7%	23.3%	0%
GPT-5.6 Luna	3.3%	72.5%	24.2%	0%
GPT-5.6 Terra	14.2%	59.2%	26.7%	0%
Claude Opus 5	66.7%	33.3%	0%	0%
Claude Sonnet 5	77.5%	20.8%	1.7%	0%

so the number of seats is the only thing deliberately varied. The two-player arm is not a near neighbour of the headline game: evaluating the group-count stage rule of the main paper at two companies returns its two-player payoff matrix exactly, at 1.0, 0.6, 2.4, 2.0, which `scripts/verify_matched_nplayer_design.py` asserts offline before the task is pushed, together with the invariance of every other mechanism field and the sharing of the horizon draw described below.

What was collected, and how. The full grid: every group size at every private-risk condition, $p_r^{\max} \in \{0.1, 0.6, 0.9\}$, ten repetitions each, on the admitted route `google/gemini-3-flash-preview`. That is 120 races and 3,696 decisions with zero parse failures. No Model Proxy identity available to us sustains the whole four-by-three grid, which needs roughly 7,800 requests, so the sweep was partitioned into the analysis unit itself, one (group size, risk) cell of ten repetitions, and each cell was collected whole on one identity so that it carries its own race-clustered interval rather than being stitched together across accounts. Cells were assigned to identities in a plan committed before the first was started, and the assignment rotates across risk levels so that no identity is confounded with a group size; every cell records its collecting identity in the artifact. Every cell's bootstrap generator is derived from a digest of that cell alone, so collecting one condition cannot move an interval already reported for another, and `scripts/analyze_nplayer_matched.py` refuses to report any risk level that is missing an arm.

Each cell was collected on a different Kaggle identity, again by a plan fixed in advance rather than by retrying until something succeeded: $N = 2$ on one account, $N = 3$, $N = 4$ and $N = 5$ on three others, each cell collected whole so that it carries its own race-clustered interval. The identity is a billing boundary rather than a model boundary, since the route, prompt, parser, protocol, temperature request and seed structure are identical across cells, but it is a difference between runs and is recorded per cell in the artifact rather than argued away.

The contrast is paired on the horizon. The game seed is base + rep and does not depend on the treatment name or on the number of seats, so one repetition index reuses a single horizon stopping-draw stream at every group size. Differencing a group size against the two-player arm inside a repetition therefore removes the horizon draw from the comparison. The analyser re-derives that pairing from the recorded game seed instead of assuming it, and it holds for all ten blocks in every contrast.

Table 21: Matched group-size comparison on `google/gemini-3-flash-preview`. Ten races per cell, zero parse failures throughout. Rates carry 95% percentile bootstrap intervals over races; the contrast is the mean difference against the two-player arm within a repetition block, with its interval taken over the ten blocks. Horizon pairing was verified for 10 of 10 blocks in every row. At $p_r^{\max} = 0.1$ every cell is at the ceiling, so its intervals have zero width: that is an absence of variation in the sample, not precision. The five-company cell at $p_r^{\max} = 0.6$ is at the ceiling too, so its contrast there is truncated, which leaves the high-risk condition as the only one that measures the five-company effect against no boundary.

N	Decisions	UNSAFE rate (95% CI)	minus N=2 (95% CI)	
Low private risk, $p_r^{\max} = 0.1$				
2	176	100.0% (100.0–100.0)	baseline	
3	264	100.0% (100.0–100.0)	+0.0 pp	(+0.0 to +0.0)
4	352	100.0% (100.0–100.0)	+0.0 pp	(+0.0 to +0.0)
5	440	100.0% (100.0–100.0)	+0.0 pp	(+0.0 to +0.0)
Medium private risk, $p_r^{\max} = 0.6$				
2	176	68.8% (65.2–72.4)	baseline	
3	264	80.3% (76.0–83.8)	+11.1 pp	(+6.3 to +15.1)
4	352	98.3% (95.6–100.0)	+28.2 pp	(+22.6 to +33.2)
5	440	100.0% (100.0–100.0)	+31.0 pp	(+26.4 to +35.3)
High private risk, $p_r^{\max} = 0.9$				
2	176	58.0% (50.7–63.7)	baseline	
3	264	68.9% (62.3–74.7)	+11.0 pp	(+5.9 to +16.4)
4	352	83.8% (80.6–86.2)	+28.1 pp	(+24.5 to +32.0)
5	440	96.1% (92.3–98.6)	+40.0 pp	(+33.2 to +47.4)

What it shows. UNSAFE play rises monotonically with the number of competing companies at every private-risk condition that leaves room for it to rise, and all six of those contrasts exclude zero. Because the two-player arm is the headline game itself, this also says something about the rest of the paper: the two-player results are the low end of that curve, not a special case beside it.

The size of the effect is what makes the grid worth having. Going from two companies to three adds 11.1 points at $p_r^{\max} = 0.6$ and 11.0 at 0.9; going to four adds 28.2 and 28.1. The two risk conditions were collected on different days and different identities, and they agree to a tenth of a point. The five-company contrast is the exception, +31.0 against +40.0, and the reason is visible in the table rather than interesting: at $p_r^{\max} = 0.6$ the five-company cell is already at 100%, so its contrast is truncated by the boundary and understates the effect. The high-risk condition is the only one that measures the five-company step against no ceiling.

The low risk condition is where the comparison becomes informative about its own limits. There every cell is at 100%, so the group-size contrast is exactly zero at every size. The right reading is not that group size stops mattering when private risk is low. It is that this route is already at the ceiling before any competitor is added, so the design has no room left in which an effect could appear. The same route reaches 98.9% and 100.0% UNSAFE at $p_r^{\max} = 0.1$ in the two independent neutral-baseline runs of §A.5, collected under a different protocol on a different engine, so the ceiling is a property of the route at low risk rather than an artifact of this sweep. Read together with the ceiling at $p_r^{\max} = 0.6$ for five companies, the pattern is that this measurement has power only where play is off the boundary, which is a statement about the instrument and not about where the mechanism operates.

What it does not show. Group size cannot be separated from everything that comes with it, and no analysis repairs that. The group-count rule makes the stage payoff depend on how many companies there are, and the state description names each of them, so a larger group is also a different payoff and a longer prompt. The estimand is therefore the effect of group size *within this game's own definition.* The comparison covers one route and ten races per cell, so it is not a claim about endpoints or about a population of models. It is regenerated by `scripts/analyze_nplayer_matched.py`, which refuses to report a risk level unless all four group sizes are present, refuses a cell with a parse failure or the wrong seat count, and records the collecting identity of every cell it uses.

A.23 The audited route against a rival it cannot influence

Every gameplay result reported elsewhere in this paper is self-play: both companies in a race are the same endpoint. That design cannot separate a route responding to its rival from a route running through a phase of its own, because the rival is the route. It also leaves this paper's two lanes disconnected. The evolutionary lane is built on four reduced strategies, *Always Safe*, *Always Unsafe*, *Conditional Safe* and *Conditional Unsafe*, and until now the empirical lane had never played against any of them.

What was collected. The route admitted with the highest score, `google/gemini-3-flash-preview`, played the same game against each of those four strategies at each of the three private-risk levels: twelve cells, ten repetitions each, 120 races and 1,116 route decisions with zero parse failures. The rival is executed by the task file rather than called, so it costs no requests and cannot be wrong about its own strategy for any reason a model could be.

Three properties of the design are load-bearing and were asserted offline before any of it was pushed. The prompt is byte-identical to the neutral baseline's and never says the rival is scripted, so a race here is comparable with a race there; the rival is a strategy, not a disclosure. The route's seat is counterbalanced, five repetitions in each seat per cell, because this study has measured a seat effect in the neutral baseline, on prompts that differ only in which of two names is whose, and a fixed seat would fold that effect into every number. And a conditional strategy answers the *route*'s previous move rather than its own, which is the property that separates a rival from a mirror: a strategy answering its own history would be a constant, and the whole conditional arm would measure nothing.

How the rival was audited. A scripted rival is code, so it cannot be audited by reading a response. It was audited by replay: `scripts/ingest_scripted_cell.py` recomputes every rival move from the route's own recorded moves and refuses the cell on a single deviation. All twelve cells replayed clean. This is the one failure that would have left the data looking perfectly healthy while answering a different question than the one asked.

Table 22: The audited route against each scripted rival. Ten races and 93 route decisions per cell, zero parse failures, both seats used in every cell. Rates carry 95% percentile bootstrap intervals over races. Contrasts are differenced inside a repetition, so the horizon stopping draw is removed from the comparison, with the interval taken over the ten blocks; the seed pairing was verified for all ten in every contrast. Ten blocks make the resampling distribution coarse, so an endpoint carries a few tenths of a point of Monte Carlo movement and is quoted to one decimal for consistency rather than because it is resolved that finely. The two cells at 100.0% sit on a boundary and their intervals have zero width for that reason.

The scripted rival	$p_r^{\max} = 0.1$	0.6	0.9
Always Safe	24.7%	17.2%	14.0%
Conditional Safe	76.3%	57.0%	57.0%
Conditional Unsafe	100.0%	74.2%	67.7%
Always Unsafe	100.0%	89.2%	88.2%
Rival unsafe minus rival safe	+73.5	+67.4	+71.7
95% CI	(67.8, 78.7)	(57.1, 76.6)	(65.6, 77.5)
The rival's opening move alone	+25.1	+15.8	+11.5
95% CI	(20.9, 29.7)	(8.3, 23.5)	(4.5, 19.7)

What it shows. The rival's stance moves this route by about seventy points, and the stated risk barely changes that number. The rate against *Always Unsafe* minus the rate against *Always Safe* is +73.5, +67.4 and +71.7 points at the three risk levels. Read down a column instead and the whole range of the rival's behaviour is worth far more than the whole range of the stated risk.

We name that contrast for what it compares rather than for a motive. Calling the safe-rival arm exploitation would assert that the route takes the opportunity a non-punishing rival offers, and the measured rates say the opposite: at 24.7%, 17.2% and 14.0% that arm carries the lowest rates anywhere in the campaign, far below what the same route plays against itself. What the route does is match its rival, keeping most of its restraint against a rival that keeps its own and abandoning nearly all of it against one that never had any.

The rival's opening move alone is worth +25.1, +15.8 and +11.5 points. *Conditional Safe* and *Conditional Unsafe* are the same strategy from round two onward, both repeating the route's previous move, so their difference isolates the first move and nothing else. It matters most where risk is cheapest, and at $p_r^{\max} = 0.1$ a rival that merely opens Unsafe and then mirrors drives the route to every one of its 93 decisions being Unsafe, the same as a rival that is Unsafe throughout. One move in round one is enough to fix the rest of the race.

What it means for the rest of this paper. A self-play rate is not a measurement of how a route treats risk. Against a fixed and safe rival this route's rates fall monotonically with the stated risk, 24.7%, 17.2%, 14.0%, which is the response the game is designed to elicit. The same route in the neutral baseline of §A.4 reads 98.9%, 73.1% and 60.2%. That is an equilibrium of two copies of one policy escalating each other, and it says more about what happens when this policy meets itself than about what the policy does with risk. Every self-play number in this paper should be read that way.

What it does not show. One route, one game, one prompt version. A scripted rival establishes what the route does against *that* rival, and four strategies are four points in a space of policies rather than a sample of opponents. The design is causal in the narrow and precise sense that the rival cannot be influenced by the route, so the direction of the effect is not in question; it is not causal about anything outside this game. Nothing here licenses a claim about strategy, intent, or reasoning: these are conditional frequencies. It is regenerated by `scripts/analyze_scripted_opponent.py`, which refuses a cell that did not complete, that carries a parse failure, that is the wrong protocol, whose seat counterbalance is not five and five, or whose rival deviated from the strategy its directory claims.

REFERENCES

[1] Gati V Aher, Rosa I. Arriaga, and Adam Tauman Kalai. 2023. Using Large Language Models to Simulate Multiple Humans and Replicate Human Subject Studies. In *Proceedings of the 40th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 202)*, Andreas Krause, Emma Brunskill, Kyunghyun Cho, Barbara Engelhardt, Sivan Sabato, and Jonathan Scarlett (Eds.). PMLR, Honolulu, Hawaii, USA, 337–371. https://proceedings.mlr.press/v202/aher23a.html

[2] Elif Akata, Lion Schulz, Julian Coda-Forno, Seong Joon Oh, Matthias Bethge, and Eric Schulz. 2025. Playing repeated games with large language models. *Nature Human Behaviour* 9, 7 (01 Jul 2025), 1380–1390. https://doi.org/10.1038/s41562-025-02172-y

[3] R. Maria del Rio-Chanona, Marco Pangallo, and Cars Hommes. 2025. Can Generative AI Agents Behave Like Humans? Evidence from Laboratory Market Experiments. arXiv:2505.07457 [econ.GN] https://arxiv.org/abs/2505.07457 Preprint.

[4] Elias Fernández Domingos and The Anh Han. 2026. Falling Behind Drives Unsafe Development in an Idealised AI Race Experiment. arXiv:2607.26034 [cs.AI] https://arxiv.org/abs/2607.26034

[5] Oliver Gürtler, Lennart Struth, and Max Thon. 2023. Competition and risk-taking. *European Economic Review* 160 (2023), 104592. https://doi.org/10.1016/j.euroecorev.2023.104592

[6] Nathan Herr, Fernando Acero, Roberta Raileanu, María Pérez-Ortiz, and Zhibin Li. 2024. Are Large Language Models Strategic Decision Makers? A Study of Performance and Bias in Two-Player Non-Zero-Sum Games. arXiv:2407.04467 [cs.AI] https://arxiv.org/abs/2407.04467

[7] Petra Nieken and Dirk Sliwka. 2010. Risk-taking tournaments – Theory and experimental evidence. *Journal of Economic Psychology* 31, 3 (2010), 254–268. https://doi.org/10.1016/j.joep.2009.03.009

[8] Serkan Ozbeklik and Janet Kiholm Smith. 2017. Risk taking in competition: Evidence from match play golf tournaments. *Journal of Corporate Finance* 44 (2017), 506–523. https://doi.org/10.1016/j.jcorpfin.2014.05.003

[9] Pouya Pezeshkpour and Estevam Hruschka. 2024. Large Language Models Sensitivity to the Order of Options in Multiple-Choice Questions. In *Findings of the Association for Computational Linguistics: NAACL 2024*. Association for Computational Linguistics, Mexico City, Mexico, 2006–2017. https://doi.org/10.18653/v1/2024.findings-naacl.130

[10] Isaac Robinson and John Burden. 2025. Framing the Game: How Context Shapes LLM Decision-Making. arXiv:2503.04840 [cs.CL] https://arxiv.org/abs/2503.04840

[11] Abel Salinas and Fred Morstatter. 2024. The Butterfly Effect of Altering Prompts: How Small Changes and Jailbreaks Affect Large Language Model Performance. arXiv:2401.03729 [cs.CL] https://arxiv.org/abs/2401.03729

[12] Melanie Sclar, Yejin Choi, Yulia Tsvetkov, and Alane Suhr. 2024. Quantifying Language Models' Sensitivity to Spurious Features in Prompt Design or: How I Learned to Start Worrying About Prompt Formatting. In *International Conference on Learning Representations*. OpenReview.net, Vienna, Austria. https://proceedings.iclr.cc/paper_files/paper/2024/hash/6c0e99d736da621403018ca7b32b1a4d-Abstract-Conference.html

[13] Sheng-Lun Wei, Cheng-Kuang Wu, Hen-Hsen Huang, and Hsin-Hsi Chen. 2024. Unveiling Selection Biases: Exploring Order and Token Sensitivity in Large Language Models. In *Findings of the Association for Computational Linguistics: ACL 2024*. Association for Computational Linguistics, Bangkok, Thailand, 5598–5621. https://doi.org/10.18653/v1/2024.findings-acl.333

[14] Zhaofeng Wu, Linlu Qiu, Alexis Ross, Ekin Akyürek, Boyuan Chen, Bailin Wang, Najoung Kim, Jacob Andreas, and Yoon Kim. 2024. Reasoning or Reciting? Exploring the Capabilities and Limitations of Language Models Through Counterfactual Tasks. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Association for Computational Linguistics, Mexico City, Mexico, 1819–1862. https://doi.org/10.18653/v1/2024.naacl-long.102

[15] Mingqian Zheng, Jiaxin Pei, Lajanugen Logeswaran, Moontae Lee, and David Jurgens. 2024. When "A Helpful Assistant" Is Not Really Helpful: Personas in System Prompts Do Not Improve Performances of Large Language Models. In *Findings of the Association for Computational Linguistics: EMNLP 2024*. Association for Computational Linguistics, Miami, Florida, USA, 15126–15154. https://doi.org/10.18653/v1/2024.findings-emnlp.888